南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

考试科目: 线 性 代 数 A　　开课单位: 数 学 系
考试时长: 120 分钟　　命题教师: 线性代数教学团队

题 号	1	2	3	4	5	6	7	8
分 值	15 分	25 分	10 分	16 分	10 分	6 分	16 分	12 分

本试卷共 (8) 大题, 满分 (110) 分. 请将所有答案写在答题本上.

This exam includes 8 questions and the score is 110 in total. Write all your answers on the examination book.

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1. (15 points, 3 points each) Multiple Choice. Only one choice is correct.

(共 15 分, 每小题 3 分) 选择题, 只有一个选项是正确的.

(1) Let A and B be $n \times n$ complex matrices. Which of the following statements is correct? ( )

(A) If A and B are diagonalizable, so is $A+B$.

(B) If A and B are diagonalizable, so is AB.

(C) If A is invertible and A^2 is diagonalizable, then A is diagonalizable.

(D) If the distinct eigenvalues of A are 1 and 0, then A is diagonalizable.

设 A 和 B 为 n 阶复方阵. 下列陈述中哪个是正确的? ( )

(A) 如果 A 和 B 都可对角化, 则 $A+B$ 也可对角化.

(B) 如果 A 和 B 都可对角化, 则 AB 也可对角化.

(C) 如果 A 为可逆且 A^2 可对角化, 则 A 也可对角化.

(D) 如果 A 的互不相同的特征值为 0 或 1, 则 A 可对角化.

(2) The equation $2x_1x_2 - 2x_1x_3 + 2x_2x_3 = 1$ represents a graph of ( )

(A) An ellipsoid.

(B) Hyperboloid of one sheet.

(C) Hyperboloid of two sheets.

(D) Hyperbolic paraboloid.

$2x_1x_2 - 2x_1x_3 + 2x_2x_3 = 1$ 表示的曲面是 ( )

(A) 椭球面.

(B) 单叶双曲面.

(C) 双叶双曲面.

(D) 双曲抛物面.

(3) Let A be a 3×3 matrix, and let B be the matrix formed by adding the second column of A to its first column. Suppose that after exchanging the second and third rows of B, the resulting matrix is the 3×3 identity matrix. Let $P_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$. Then $A =$ (　　)

(A) P_1P_2.
(B) $P_1^{-1}P_2$.
(C) P_2P_1.
(D) $P_2P_1^{-1}$.

设 A 为 3 阶方阵, 将 A 的第二列加到第一列得矩阵 B. 假设交换 B 的第二行与第三行可以得到 3 阶单位矩阵. 记 $P_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$. 则 $A =$ (　　)

(A) P_1P_2.
(B) $P_1^{-1}P_2$.
(C) P_2P_1.
(D) $P_2P_1^{-1}$.

(4) Let $A = \begin{bmatrix} -1 & 2 & 3 \\ -3 & 6 & 8 \\ 2 & -4 & t \end{bmatrix}$, where $t \in \mathbb{R}$. Suppose $\operatorname{rank}(A) = 2$. Then (　　)

(A) $t = -6$.
(B) $t = 6$.
(C) $t \neq 0$.
(D) t can be any real number.

设 $A = \begin{bmatrix} -1 & 2 & 3 \\ -3 & 6 & 8 \\ 2 & -4 & t \end{bmatrix}$, 其中 $t \in \mathbb{R}$. 假设 $\operatorname{rank}(A) = 2$. 则 (　　)

(A) $t = -6$.
(B) $t = 6$.
(C) $t \neq 0$.
(D) t 可以是任意实数.

(5) Which of the following statements is incorrect ? (　　)

(A) For any matrix A, $\operatorname{rank}(A) = \dim C(A)$.

(B) If $v_1, \cdots, v_m$ are pairwise orthogonal nonzero vectors, then the vectors $v_1, \cdots, v_m$ are linear independent.

(C) If A is an upper triangular $n \times n$ matrix such that $A^2 = 0$, then $A = 0$.

(D) Let A, B be $n \times n$ matrices such that AB is invertible. Then both A and B are invertible.

下列哪个论断是错误的？(　　)

(A) 对于任意矩阵 A, $\operatorname{rank}(A) = \dim C(A)$.

(B) 如果 $v_1, \cdots, v_m$ 是一组两两正交的非零向量, 则向量组 $v_1, \cdots, v_m$ 线性无关.

(C) 如果 A 是 $n \times n$ 上三角矩阵且 $A^2 = 0$, 则 $A = 0$.

(D) 设 A, B 为 $n \times n$ 矩阵且 AB 可逆. 则 A 和 B 都可逆.

2. (25 points, 5 points each) Fill in the blanks. (共 25 分, 每小题 5 分) 填空题.

(1) Let A, B be invertible $n \times n$ matrices. Then the inverse of the block matrix $\begin{bmatrix} 0 & A \\ B & 0 \end{bmatrix}$ is __________

设 A, B 均为 $n \times n$ 可逆矩阵. 则分块矩阵 $\begin{bmatrix} 0 & A \\ B & 0 \end{bmatrix}$ 的逆为 __________

(2) Suppose A is a 3×4 matrix and $\dim N(A) = 2$. Then $\dim N(A^T) =$ __________

设 A 为 3×4 矩阵且 $\dim N(A) = 2$. 则 $\dim N(A^T) =$ __________

(3) Let $A = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix}$. Then $A^{-1} =$__________

设 $A = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix}$. 则 $A^{-1} =$__________

(4) Let u, v be vectors in $\mathbb{R}^n$ such that $\|u\| = 3$, $\|v\| = 4$ and $u^T v = -3$.
Then $\|2u + 3v\| =$__________

设 u, v 为 $\mathbb{R}^n$ 中的向量, 满足 $\|u\| = 3$, $\|v\| = 4$ 以及 $u^T v = -3$. 则 $\|2u+3v\| =$__________

(5) Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & -1 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}$.

Then the least squares solution to $Ax = b$ is $\hat{x} =$__________

设 $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & -1 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}$.

则 $Ax = b$ 的最小二乘解是 $\hat{x} =$__________

3. (10 points) Let

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \ v_2 = \begin{bmatrix} 3 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \ v_3 = \begin{bmatrix} 3 \\ 1 \\ 2 \\ 1 \end{bmatrix}.$$

(a) Find an orthonormal basis, q_1, q_2, q_3, for the subspace V of $\mathbb{R}^4$ spanned by the column vectors v_1, v_2, v_3.

(b) Let

$$A = \begin{bmatrix} 1 & 3 & 3 \\ 0 & 0 & 1 \\ 1 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}, \ b = \begin{bmatrix} 2020 \\ 2021 \\ 2022 \\ 2023 \end{bmatrix}.$$

Find the least-squares solution of $Ax = b$.

假定

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \ v_2 = \begin{bmatrix} 3 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \ v_3 = \begin{bmatrix} 3 \\ 1 \\ 2 \\ 1 \end{bmatrix}.$$

(a) 设 V 为由 v_1, v_2, v_3 生成的 $\mathbb{R}^4$ 的子空间，求 V 的一组标准正交基.

(b) 记

$$A = \begin{bmatrix} 1 & 3 & 3 \\ 0 & 0 & 1 \\ 1 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}, \ b = \begin{bmatrix} 2020 \\ 2021 \\ 2022 \\ 2023 \end{bmatrix}.$$

求 $Ax = b$ 在最小二乘意义下的解.

4. (16 points) Compute the nth order determinant:

$$|A| = \begin{vmatrix} a & 0 & \cdots & \cdots & 0 & 1 \\ 0 & a & 0 & \cdots & \cdots & 0 \\ \vdots & 0 & \ddots & \ddots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \cdots & \cdots & 0 & a & 0 \\ 1 & 0 & \cdots & \cdots & 0 & a \end{vmatrix}, n \geq 2.$$

(16 分) 计算如下的 n 阶行列式:

$$|A| = \begin{vmatrix} a & 0 & \cdots & \cdots & 0 & 1 \\ 0 & a & 0 & \cdots & \cdots & 0 \\ \vdots & 0 & \ddots & \ddots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \cdots & \cdots & 0 & a & 0 \\ 1 & 0 & \cdots & \cdots & 0 & a \end{vmatrix}, n \geq 2.$$

5. (10 points) Let $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 4 & -2 \\ -3 & -3 & a \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & b \end{bmatrix}$, and A is similar to B. Find an orthogonal matrix Q, such that $Q^{-1}AQ = B$.

(10 分) Let $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 4 & -2 \\ -3 & -3 & a \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & b \end{bmatrix}$, and A is similar to B. Find an orthogonal matrix Q, such that $Q^{-1}AQ = B$.

6. (6 points) Let A, B be $n \times n$ matrices. Suppose A and B are both symmetric. Is AB necessarily symmetric? If yes, please give a proof. Otherwise please give a counterexample.

(6 分) 设 A, B 均为 $n \times n$ 矩阵. 假设 A 和 B 都是对称矩阵. AB 是否一定是对称矩阵? 若是, 请给出证明. 否则请给出一个反例.

7. (16 points)

(16 分)

8. (12 points) Let A be a 3×3 matrix such that $\text{rank}(A) = 2$ and $A^3 = 0$.

(a) Prove that $\text{rank}(A^2) = 1$.

(b) Let $\alpha_1 \in \mathbb{R}^3$ be a nonzero vector such that $A\alpha_1 = 0$. Prove that there exist vectors α_2, α_3 such that $A\alpha_2 = \alpha_1$, $A^2\alpha_3 = \alpha_1$.

(c) For any vectors α_2, α_3 described as above, show that $\alpha_1, \alpha_2, \alpha_3$ are linearly independent.

(In this problem, you are allowed to assume the statements of some questions to answer subsequent questions.)

(12 分) 设 A 是 3×3 矩阵, 它满足 $\operatorname{rank}(A)=2$ 及 $A^3=0$.

(a) 证明 $\operatorname{rank}(A^2)=1$.

(b) 设 $\alpha_1\in\mathbb{R}^3$ 是满足 $A\alpha_1=0$ 的非零向量. 证明: 存在向量 α_2,α_3 使得 $A\alpha_2=\alpha_1$, $A^2\alpha_3=\alpha_1$.

(c) 证明: 对于任意满足上述条件的向量 α_2,α_3, 向量组 $\alpha_1,\alpha_2,\alpha_3$ 线性无关.

(本题中, 允许承认前面小题的结果来用于后续问题的解答.)

1. (15 points, 3 points each) Multiple Choice. Only one choice is correct.

(共 15 分, 每小题 3 分) 选择题, 只有一个选项是正确的.

(1) Let $A = \begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix}$ and $f(t) = 1 + 2t + t^2 + t^4 - 5t^8$. Then $f(A) =$

(A) $\begin{bmatrix} 1 & 6 \\ 0 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 8 \\ 1 & 0 \end{bmatrix}$

(C) $\begin{bmatrix} 6 & 1 \\ 1 & 0 \end{bmatrix}$

(D) $\begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix}$

设 $A = \begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix}$, 且 $f(t) = 1 + 2t + t^2 + t^4 - 5t^8$, 则 $f(A) =$

(A) $\begin{bmatrix} 1 & 6 \\ 0 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 8 \\ 1 & 0 \end{bmatrix}$

(C) $\begin{bmatrix} 6 & 1 \\ 1 & 0 \end{bmatrix}$

(D) $\begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix}$

(2) Let A and B be invertible matrices. If A is similar to B, which of the following statements is **NOT** correct?

(A) A^T is similar to B^T.

(B) A^{-1} is similar to B^{-1}.

(C) $A + A^T$ is similar to $B + B^T$.

(D) $A + A^{-1}$ is similar to $B + B^{-1}$.

假定 A 和 B 都是可逆矩阵, 且 A 和 B 相似, 下列陈述中哪个是不正确的？

(A) A^T 和 B^T 相似.

(B) A^{-1} 和 B^{-1} 相似.

(C) $A + A^T$ 和 $B + B^T$ 相似.

(D) $A + A^{-1}$ 和 $B + B^{-1}$ 相似.

(3) Let $A = \begin{bmatrix} -1 & -2 & 3 \\ -2 & 6 & 8 \\ 3 & 8 & 5 \end{bmatrix}$. Then the number of positive eigenvalues of A is

(A) 0.

(B) 1.

(C) 2.

(D) 3.

设 $A = \begin{bmatrix} -1 & -2 & 3 \\ -2 & 6 & 8 \\ 3 & 8 & 5 \end{bmatrix}$, 则矩阵 A 的正的特征值的个数为

(A) 0.

(B) 1.

(C) 2.

(D) 3.

(4) The equation $2x_1x_2 - 2x_1x_3 + 2x_2x_3 = 1$ represents a graph of

(A) An ellipsoid.

(B) Hyperboloid of one sheet.

(C) Hyperboloid of two sheets.

(D) Hyperbolic paraboloid.

$2x_1x_2 - 2x_1x_3 + 2x_2x_3 = 1$ 表示的曲面是

(A) 椭球面.

(B) 单叶双曲面.

(C) 双叶双曲面.

(D) 双曲抛物面.

(5) Which of the following statements is correct?

(A) If A is a Hermitian matrix, and $x^H Ax = 0$ for all complex vectors x, then $A = O$, where O denotes the zero matrix.

(B) An $n \times n$ matrix with real eigenvalues and n linearly independent real eigenvectors is symmetric.

(C) If A is a complex matrix, and $A^T = A$, then A is diagonalizable.

(D) Let A, B be $n \times n$ real matrices, then $\det(A + B) = \det A + \det B$.

下面的哪个陈述是正确的?

(A) 如果 A 是厄密特矩阵, 而且对所有的复向量 x 都有 $x^H Ax = 0$, 那么 $A = O$, 这里 O 表示零矩阵.

(B) 一个 n 阶的方阵的所有特征值和 n 个线性无关的特征向量都是实的, 则这个矩阵是对称的.

(C) 如果 A 是一个复矩阵, 且满足 $A^T = A$, 则 A 是可对角化的.

(D) 设 A, B 都是 n 阶实方阵, 则 $\det(A+B) = \det A + \det B$.

ANS: (1) A　(2) C　(3) C　(4) B　(5) A

2. (25 points, 5 points each) Fill in the blanks. (共 25 分, 每小题 5 分) 填空题.

(1) Suppose A is a 5×4 real matrix with 3 linearly independent columns. The dimension of the row space of A is __________, and the dimension of the left nullspace of A is __________.

设一个 5×4 的实矩阵 A 有三个线性无关的列向量, 则 A 的行空间的维数为 __________, A 的左零空间的维数为 __________.

(2) If the real quadratic form $f(x_1, x_2, x_3) = a(x_1^2 + x_2^2 + x_3^2) + 4x_1x_2 + 4x_1x_3 + 4x_2x_3$ is changed to standard form $f = 6y_1^2$ by orthogonal transformation $x = Qy$, then $a =$ __________.

如果实二次型 $f(x_1, x_2, x_3) = a(x_1^2 + x_2^2 + x_3^2) + 4x_1x_2 + 4x_1x_3 + 4x_2x_3$ 经过正交变换 $x = Qy$ 化为标准形 $f = 6y_1^2$, 则 $a =$ __________.

(3) The eigenvalues of $I_3 - uv^T$ are __________. Where I_3 is the 3×3 identity matrix, and u and v are nonzero vectors in $\mathbb{R}^3$.

矩阵 $I_3 - uv^T$ 的特征值为 __________. 这里 I_3 表示 3 阶单位阵, u 和 v 是 $\mathbb{R}^3$ 中的非零向量.

(4) If $A^2 = A$ and rank $(A) = r$, then trace $(A) =$__________.

如果 $A^2 = A$ 且 rank $(A) = r$, 则 trace $(A) =$__________.

(5) Let $\alpha_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 2 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 2 \\ 1 \\ 1 \\ a \end{bmatrix}.$

If the dimension of the vector space generated by $\alpha_1, \alpha_2, \alpha_3$ is 2, then $a =$__________.

设 $\alpha_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 2 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 2 \\ 1 \\ 1 \\ a \end{bmatrix}.$

如果由 $\alpha_1, \alpha_2, \alpha_3$ 生成的子空间的维数为 2, 则 $a =$__________.

ANS: (1) 3,2　(2) 2　(3) $1, 1 - u^T v$　(4) r　(5) 6

3. (12 points) Let

$$A=\begin{bmatrix}1&3&3\\0&0&1\\1&1&2\\0&0&1\end{bmatrix}.$$

(a) Find an orthonormal basis for the column space of A;

(b) Write A as QR, where Q has orthonormal columns and R is upper triangular.

(12 分) 设

$$A=\begin{bmatrix}1&3&3\\0&0&1\\1&1&2\\0&0&1\end{bmatrix}.$$

(a) 求 A 的列空间的一组标准正交基;

(b) 将 A 分解成 QR, 其中 Q 的列是标准正交的向量, R 是上三角矩阵.

Solution.

(a) An orthonormal basis for the column space of A is:

$$q_1=\begin{bmatrix}\frac{1}{\sqrt{2}}\\0\\\frac{1}{\sqrt{2}}\\0\end{bmatrix}, q_2=\begin{bmatrix}\frac{1}{\sqrt{2}}\\0\\-\frac{1}{\sqrt{2}}\\0\end{bmatrix}, q_3=\begin{bmatrix}0\\\frac{1}{\sqrt{2}}\\0\\\frac{1}{\sqrt{2}}\end{bmatrix},$$

(b) The QR factorization of A is as follows:

$$A=QR=\begin{bmatrix}\frac{1}{\sqrt{2}}&\frac{1}{\sqrt{2}}&0\\0&0&\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}&-\frac{1}{\sqrt{2}}&0\\0&0&\frac{1}{\sqrt{2}}\end{bmatrix}\begin{bmatrix}\sqrt{2}&2\sqrt{2}&\frac{5\sqrt{2}}{2}\\0&\sqrt{2}&\frac{1}{\sqrt{2}}\\0&0&\sqrt{2}\end{bmatrix}.$$

4. (10 points) Compute the determinant of an $n \times n$ matrix A:

$$|A| = \begin{vmatrix} a & 0 & \cdots & \cdots & 0 & 1 \\ 0 & a & 0 & \cdots & \cdots & 0 \\ \vdots & 0 & \ddots & \ddots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \cdots & \cdots & 0 & a & 0 \\ 1 & 0 & \cdots & \cdots & 0 & a \end{vmatrix}, \quad n \geq 2.$$

(10 分) 计算 n 阶矩阵 A 的行列式:

$$|A| = \begin{vmatrix} a & 0 & \cdots & \cdots & 0 & 1 \\ 0 & a & 0 & \cdots & \cdots & 0 \\ \vdots & 0 & \ddots & \ddots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \cdots & \cdots & 0 & a & 0 \\ 1 & 0 & \cdots & \cdots & 0 & a \end{vmatrix}, \quad n \geq 2.$$

Solution. $a^n - a^{n-2}$.

4. (10 points) Compute the determinant of an $n \times n$ matrix A:

$$|A| = \begin{vmatrix} a & 0 & \cdots & \cdots & 0 & 1 \\ 0 & a & 0 & \cdots & \cdots & 0 \\ \vdots & 0 & \ddots & \ddots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \cdots & \cdots & 0 & a & 0 \\ 1 & 0 & \cdots & \cdots & 0 & a \end{vmatrix}, \quad n \geq 2.$$

(10 分) 计算 n 阶矩阵 A 的行列式:

$$|A| = \begin{vmatrix} a & 0 & \cdots & \cdots & 0 & 1 \\ 0 & a & 0 & \cdots & \cdots & 0 \\ \vdots & 0 & \ddots & \ddots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \cdots & \cdots & 0 & a & 0 \\ 1 & 0 & \cdots & \cdots & 0 & a \end{vmatrix}, \quad n \geq 2.$$

Solution. $a^n - a^{n-2}$.

6. (12 points) Let

$$A = \begin{bmatrix} 3 & -3 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}$$

(a) Find all the singular values of A;

(b) Find a singular value decomposition of A.

(12 分) 设

$$A = \begin{bmatrix} 3 & -3 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}$$

(a) 求 A 的所有奇异值;

(b) 求矩阵 A 的一个奇异值分解.

Solution.

(a) The singular values of A are $3\sqrt{2}, \sqrt{2}$;

(b) A singular value decomposition of A is as follows:

$$A = U\Sigma V^T = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 3\sqrt{2} & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

8. (8 points) Let A be an $n \times n$ real symmetric positive definite matrix, and B be an $n \times n$ real symmetric positive semidefinite matrix.

   (a) Prove that the eigenvalues of AB are all nonnegative real numbers.

   (b) Prove that AB is diagonalizable.

   (8 分) 设 A 为 n 阶正定实对称矩阵, B 为 n 阶半正定实对称矩阵.

   (a) 证明: AB 的所有特征值都是非负实数.

   (b) 证明: AB 可对角化.

Solution.

(a) Since A is positive definite, and B is positive semidefinite, then we can find P and Q such that

$$A = P^T P, B = Q^T Q,$$

where P is an invertible matrix. It follows that

$$AB = P^T P Q^T Q = P^T P Q^T Q P^T (P^T)^{-1}.$$

This means that AB is similar to $PQ^T QP^T$. This together with the fact that $PQ^T QP^T$ is a positive semidefinite matrix complete the proof.

(b) Since A is positive definite, then there is an invertible matrix C such that

$$C^T AC = I_n,\ C^T AB(C^T)^{-1} = C^T ACC^{-1}B(C^T)^{-1}.$$

$M = C^{-1}B(C^T)^{-1}$ is a positive semidefinite matrix, therefore we can find an orthogonal matrix Q such that

$$Q^T MQ = \begin{bmatrix} \Lambda & 0 \\ 0 & 0 \end{bmatrix}.$$

Therefore

$$Q^T C^T AB(C^T)^{-1}Q = Q^T C^T ACC^{-1}B(C^T)^{-1}Q = Q^T MQ = \begin{bmatrix} \Lambda & 0 \\ 0 & 0 \end{bmatrix}.$$

It follows that AB is diagonalizable.

南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

考试科目：线性代数 A　　开课单位：数学系
考试时长：120 分钟　　命题教师：线性代数教学团队

题号	1	2	3	4	5	6	7	8
分值	15 分	25 分	12 分	10 分	12 分	12 分	8 分	6 分

本试卷共 (8) 大题, 满分 (100) 分. 请将所有答案写在答题本上.

This exam includes 8 questions and the score is 100 in total. Write all your answers on the examination book.

本套试卷为 B 卷　Version B

1. (15 points, 3 points each) Multiple Choice. Only one choice is correct.

(共 15 分, 每小题 3 分) 选择题, 只有一个选项是正确的.

(1) Let $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 0 & 1 & 1 \end{bmatrix}$ and let $\alpha_1, \alpha_2, \alpha_3$ be linearly independent column vectors in $\mathbb{R}^3$. Then the rank of the vector system $A\alpha_1, A\alpha_2, A\alpha_3$ (　)

(A) must be 1.

(B) must be 2.

(C) must be 3.

(D) can be 1 or 2.

设 $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 0 & 1 & 1 \end{bmatrix}$, $\alpha_1, \alpha_2, \alpha_3$ 为 $\mathbb{R}^3$ 中线性无关的向量组. 则向量组 $A\alpha_1, A\alpha_2, A\alpha_3$ 的秩 (　)

(A) 一定是 1.

(B) 一定是 2.

(C) 一定是 3.

(D) 可能是 1 也可能是 2.

(2) Let I_n be the identity matrix of order n and let α be a column vector of length 1 in $\mathbb{R}^n$. Then (　)

(A) $I_n - \alpha\alpha^T$ is not invertible.

(B) $I_n + \alpha\alpha^T$ is not invertible.

(C) $I_n + 2\alpha\alpha^T$ is not invertible.

(D) $I_n - 2\alpha\alpha^T$ is not invertible.

设 I_n 为 n 阶单位矩阵, α 是 $\mathbb{R}^n$ 中长度为 1 的列向量. 则 (　　)

(A) $I_n - \alpha\alpha^T$ 不可逆.

(B) $I_n + \alpha\alpha^T$ 不可逆.

(C) $I_n + 2\alpha\alpha^T$ 不可逆.

(D) $I_n - 2\alpha\alpha^T$ 不可逆.

(3) Let $M = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}$. Which of the following matrices is congruent to M? (　　)

(A) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

(B) $B = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$

(C) $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

(D) $A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$

设 $M = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}$. 下列哪个矩阵与 M 相合 (也称合同)? (　　)

(A) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

(B) $B = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$

(C) $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

(D) $A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$

(4) Let $\alpha_1, \alpha_2, \alpha_3$ be column vectors in $\mathbb{R}^3$ such that the matrix $P = (\alpha_1, \alpha_2, \alpha_3)$ is invertible. Suppose A is a 3×3 matrix such that $P^{-1}AP = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$. Then $A(\alpha_1 + \alpha_2 + \alpha_3) =$ (　　)

(A) $\alpha_1 + \alpha_2$.

(B) $\alpha_2 + 3\alpha_3$.

(C) $\alpha_2 + \alpha_3$.

(D) $\alpha_1 + 2\alpha_3$.

设 $\alpha_1, \alpha_2, \alpha_3$ 为 $\mathbb{R}^3$ 中的列向量, 使得矩阵 $P = (\alpha_1, \alpha_2, \alpha_3)$ 可逆. 假设 A 为 3×3 矩阵, 满足 $P^{-1}AP = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$. 则 $A(\alpha_1+\alpha_2+\alpha_3) = (\quad)$

(A) $\alpha_1 + \alpha_2$.

(B) $\alpha_2 + 2\alpha_3$.

(C) $\alpha_2 + \alpha_3$.

(D) $\alpha_1 + 2\alpha_3$.

(5) Which of the following statements is true? (　　)

(A) Let A be an $m\times n$ complex matrix and $B = A^H A$. (Here $A^H = \overline{A}^T$ denotes the conjugate transpose of A.) Then the matrix $B + iI_n$ is invertible.

(B) If A is a real square matrix of order n and $x^T A x = 0$ for all $x \in \mathbb{R}^n$, then A must be the zero matrix.

(C) Suppose A and B are invertible square matrices. If A and B are similar to each other, then $A + A^T$ is similar to $B + B^T$.

(D) If A is an upper triangular (square) matrix and A is similar to a diagonal matrix, then A must be a diagonal matrix.

下列哪个论断是正确的? (　　)

(A) 设 A 为 $m\times n$ 复矩阵, $B = A^H A$. (这里 $A^H = \overline{A}^T$ 表示 A 的共轭转置.) 则矩阵 $B + iI_n$ 可逆.

(B) 如果 A 是 n 阶实方阵, 且对任意 $x \in \mathbb{R}^n$ 均有 $x^T A x = 0$, 则 A 一定是零矩阵.

(C) 假设 A 和 B 都是可逆方阵. 如果 A 和 B 相似, 则 $A + A^T$ 和 $B + B^T$ 相似.

(D) 若 A 是上三角(方)阵并且 A 相似于某个对角阵, 则 A 一定是对角阵.

2. (25 points, 5 points each) Fill in the blanks. (共 25 分, 每小题 5 分) 填空题.

(1) Let A be a 5×4 matrix of rank 3. Then the dimension of its left null space $N(A^T)$ is __________

设 A 为秩等于 3 的 5×4 矩阵. 则它的左零空间 $N(A^T)$ 维数是 __________

(2) Let A be a square matrix of order 3 and let $\alpha_1, \alpha_2, \alpha_3$ be linearly independent vectors in $\mathbb{R}^3$. Suppose $A\alpha_1 = 2\alpha_1 + \alpha_2 + \alpha_3$, $A\alpha_2 = \alpha_2 + 2\alpha_3$ and $A\alpha_3 = -\alpha_2 + \alpha_3$. Then the real eigenvalues of A are __________

设 A 为 3 阶方阵, $\alpha_1, \alpha_2, \alpha_3$ 为 $\mathbb{R}^3$ 中的线性无关向量组. 假设 $A\alpha_1 = 2\alpha_1 + \alpha_2 + \alpha_3$, $A\alpha_2 = \alpha_2 + 2\alpha_3$ 且 $A\alpha_3 = -\alpha_2 + \alpha_3$. 则 A 的实特征值为 __________

(3) Let A, B be square matrices of order n. Suppose $|A| = 3$, $|B| = 2$ and $|A^{-1} + B| = 2$. Then $|A + B^{-1}| =$__________

设 A, B 为 n 阶方阵. 假设 $|A| = 3$, $|B| = 2$ 而 $|A^{-1}+B| = 2$. 则 $|A+B^{-1}| =$__________

(4) Let ξ_1, ξ_2, ξ_3 be a basis of $\mathbb{R}^3$, and let $\eta_1 = \xi_1$, $\eta_2 = \xi_1+\xi_2$, $\eta_3 = \xi_3-\xi_1$. Let $\mathrm{Id} : \mathbb{R}^3 \to \mathbb{R}^3$ be the identity map.

Then the matrix A representing Id in the basis η_1, η_2, η_3 is ___________

设 ξ_1, ξ_2, ξ_3 为 $\mathbb{R}^3$ 的一组基, $\eta_1 = \xi_1$, $\eta_2 = \xi_1 + \xi_2$, $\eta_3 = \xi_3 - \xi_1$. 记 $\mathrm{Id} : \mathbb{R}^3 \to \mathbb{R}^3$ 为恒等映射. 则 Id 在 η_1, η_2, η_3 这组基下的矩阵表示是 $A =$___________

(5) Let $a \in \mathbb{R}$ and let $C \subseteq \mathbb{R}^2$ be the zero locus of the equation $3x^2 - 2axy + 3y^2 - 1 = 0$. Determine for which values of a the curve C is an ellipse. (A circle is also considered as an ellipse.) ___________

设 $a \in \mathbb{R}$, 记 $C \subseteq \mathbb{R}^2$ 为方程 $3x^2 - 2axy + 3y^2 - 1 = 0$ 的零点集. 求 a 的取值范围使得曲线 C 是个椭圆. (注意圆也认为是椭圆.) ___________

3. (12 points)

(a) Find an orthonormal basis for the column space of

$$A = \begin{bmatrix} 1 & -2 & -1 \\ 2 & 0 & 1 \\ 2 & -4 & 2 \\ 4 & 0 & 0 \end{bmatrix}.$$

(b) Write A as QR, where Q has orthonormal columns and R is upper triangular.

(12 分)

(a) 找出下列矩阵的列空间的一组标准正交 (也称规范正交) 基:

$$A = \begin{bmatrix} 1 & -2 & -1 \\ 2 & 0 & 1 \\ 2 & -4 & 2 \\ 4 & 0 & 0 \end{bmatrix}.$$

(b) 将 A 分解成乘积 QR 的形式, 其中 Q 的列是标准正交 (也称规范正交) 的向量, R 是上三角矩阵.

4. (10 points) Let $x, y \in \mathbb{R}$, $n \geq 2$ and consider the following determinant of order n:

$$D_n(x, y) = \begin{vmatrix} x+y & xy & 0 & \dots & 0 & 0 \\ 1 & x+y & xy & \dots & 0 & 0 \\ 0 & 1 & x+y & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 1 & x+y \end{vmatrix}$$

For $n = 2$, we consider $D_2(x, y)$ as $\begin{vmatrix} x+y & xy \\ 1 & x+y \end{vmatrix}$.

(a) Find a recurrence relation relating $D_n(x,y)$ to $D_{n-1}(x,y)$ and $D_{n-2}(x,y)$ for $n \geq 4$.

(b) Compute the determinant $D_n(x,y)$ for all $n \geq 2$.

(10 分) 设 $x, y \in \mathbb{R}$, $n \geq 2$. 考虑以下 n 阶行列式:

$$D_n(x,y) = \begin{vmatrix} x+y & xy & 0 & \dots & 0 & 0 \\ 1 & x+y & xy & \dots & 0 & 0 \\ 0 & 1 & x+y & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 1 & x+y \end{vmatrix}$$

当 $n=2$ 时, 我们认为 $D_2(x,y)$ 是 $\begin{vmatrix} x+y & xy \\ 1 & x+y \end{vmatrix}$.

(a) 在 $n \geq 4$ 时找出能将 $D_n(x,y)$ 和 $D_{n-1}(x,y), D_{n-2}(x,y)$ 建立联系的一个递推关系式.

(b) 对任意 $n \geq 2$ 计算行列式 $D_n(x,y)$.

5. (12 points) Let $a, b \in \mathbb{R}$ and put

$$A = \begin{bmatrix} 2 & -1 & 2 \\ 5 & a & 3 \\ -1 & b & -2 \end{bmatrix}.$$

Suppose that $p = (1,1,-1)^T$ is an eigenvector of A.

(a) Find the values of a, b and find the eigenvalue λ corresponding to the eigenvector p.

(b) Is the matrix A diagonalizable? Please explain your answer.

(12 分) 设 $a, b \in \mathbb{R}$, 令

$$A = \begin{bmatrix} 2 & -1 & 2 \\ 5 & a & 3 \\ -1 & b & -2 \end{bmatrix}.$$

假设 $p = (1,1,-1)^T$ 是 A 的一个特征向量.

(a) 求 a, b 的值并找出特征向量 p 对应的特征值 λ.

(b) 矩阵 A 是否可对角化? 请解释你的答案.

6. (12 points) Consider the ternary quadratic form

$$f(x_1, x_2, x_3) = (x_1 - x_2 + x_3)^2 + (x_2 + x_3)^2 + (x_1 + a x_3)^2$$

where $a \in \mathbb{R}$ is a parameter.

(a) What are the possible values of a if the quadratic form f is positive definite?

(b) What are the possible values of a if the equation $f(x_1, x_2, x_3) = 0$ has infinitely many solutions?

(c) Let $y = (y_1, y_2, y_3)^T$ be a new system of variables. Suppose that $f(x_1, x_2, x_3) = 0$ has infinitely many solutions.

Find an invertible linear transformation $y = Px$, where $x = (x_1, x_2, x_3)^T$, such that in the variables y_1, y_2, y_3 the quadratic form f has a diagonal form.

(12 分) 考虑三元二次型

$$f(x_1, x_2, x_3) = (x_1 - x_2 + x_3)^2 + (x_2 + x_3)^2 + (x_1 + ax_3)^2$$

其中 $a \in \mathbb{R}$ 为参数.

(a) 如果 f 是正定的, a 的取值范围是什么？

(b) 如果方程 $f(x_1, x_2, x_3) = 0$ 有无穷多个解, a 的取值可能是什么？

(c) 设 $y = (y_1, y_2, y_3)^T$ 为一组新的变元. 假设 $f(x_1, x_2, x_3) = 0$ 有无穷多个解.

找出一个可逆线性变换 $y = Px$, 其中 $x = (x_1, x_2, x_3)^T$, 使得在 y_1, y_2, y_3 这组变元下二次型 f 可以表示为对角形式.

7. (8 points) Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & -1 \end{bmatrix}$.

(a) Find all the singular values of A.

(b) Find the singular value decomposition of A, in other words, find two orthogonal matrices U and V (of suitable size) such that $A = U\Sigma V^T$.

(8 分) 令 $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & -1 \end{bmatrix}$.

(a) 求 A 的所有奇异值.

(b) 求 A 的奇异值分解. 即, 找出两个 (适当大小的) 正交矩阵 U 和 V 使得 $A = U\Sigma V^T$.

8. (6 points) Find a real 4×4 orthogonal matrix A such that A has no real eigenvalues but both A^2 and A^3 have real eigenvalues. Please explain why the matrix you give has the required properties.

(6 分) 给出一个 4×4 实正交矩阵 A 使得 A 没有实特征值, 而 A^2 和 A^3 都有实特征值. 请解释为什么你给出的矩阵满足要求.

南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

考试科目: 线性代数 A　　　　开课单位: 数 学 系
考试时长: 120 分钟　　　　命题教师: 线性代数教学团队

题 号	1	2	3	4	5	6	7	8
分 值	15 分	25 分	12 分	10 分	12 分	12 分	8 分	6 分

本试卷共 (8) 大题, 满分 (100) 分. 请将所有答案写在答题本上.

This exam includes 8 questions and the score is 100 in total. Write all your answers on the examination book.

本套试卷为 B 卷 Version B

1. (15 points, 3 points each) Multiple Choice. Only one choice is correct.

(共 15 分, 每小题 3 分) 选择题, 只有一个选项是正确的.

(1) Let $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 0 & 1 & 1 \end{bmatrix}$ and let $\alpha_1, \alpha_2, \alpha_3$ be linearly independent column vectors in $\mathbb{R}^3$. Then the rank of the vector system $A\alpha_1, A\alpha_2, A\alpha_3$ (　　)

(A) must be 1.

(B) must be 2.

(C) must be 3.

(D) can be 1 or 2.

设 $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 0 & 1 & 1 \end{bmatrix}$, $\alpha_1, \alpha_2, \alpha_3$ 为 $\mathbb{R}^3$ 中线性无关的向量组. 则向量组 $A\alpha_1, A\alpha_2, A\alpha_3$ 的秩 (　　)

(A) 一定是 1.

(B) 一定是 2.

(C) 一定是 3.

(D) 可能是 1 也可能是 2.

(2) Let I_n be the identity matrix of order n and let α be a column vector of length 1 in $\mathbb{R}^n$. Then (　　)

(A) $I_n - \alpha\alpha^T$ is not invertible.

(B) $I_n + \alpha\alpha^T$ is not invertible.

(C) $I_n + 2\alpha\alpha^T$ is not invertible.

(D) $I_n - 2\alpha\alpha^T$ is not invertible.

设 I_n 为 n 阶单位矩阵, α 是 $\mathbb{R}^n$ 中长度为 1 的列向量. 则 (　　)

(A) $I_n - \alpha\alpha^T$ 不可逆.

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(C) $I_n + 2\alpha\alpha^T$ 不可逆.

(D) $I_n - 2\alpha\alpha^T$ 不可逆.

(3) Let $M = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}$. Which of the following matrices is congruent to M ? (　　)

(A) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

(B) $B = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$

(C) $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

(D) $A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$

设 $M = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}$. 下列哪个矩阵与 M 相合 (也称合同)? (　　)

(A) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

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(C) $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

(D) $A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$

(4) Let $\alpha_1, \alpha_2, \alpha_3$ be column vectors in $\mathbb{R}^3$ such that the matrix $P = (\alpha_1, \alpha_2, \alpha_3)$ is invertible. Suppose A is a 3×3 matrix such that $P^{-1}AP = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$. Then $A(\alpha_1 + \alpha_2 + \alpha_3) =$ (　　)

(A) $\alpha_1 + \alpha_2$.

(B) $\alpha_2 + 3\alpha_3$.

(C) $\alpha_2 + \alpha_3$.

(D) $\alpha_1 + 2\alpha_3$.

设 $\alpha_1, \alpha_2, \alpha_3$ 为 $\mathbb{R}^3$ 中的列向量, 使得矩阵 $P=(\alpha_1, \alpha_2, \alpha_3)$ 可逆. 假设 A 为 3×3 矩阵, 满足 $P^{-1}AP=\begin{bmatrix}0&0&0\\0&1&0\\0&0&2\end{bmatrix}$. 则 $A(\alpha_1+\alpha_2+\alpha_3)=($ $)$

(A) $\alpha_1+\alpha_2$.

(B) $\alpha_2+2\alpha_3$.

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(5) Which of the following statements is true? ( )

(A) Let A be an $m\times n$ complex matrix and $B=A^HA$. (Here $A^H=\overline{A}^T$ denotes the conjugate transpose of A.) Then the matrix $B+iI_n$ is invertible.

(B) If A is a real square matrix of order n and $x^TAx=0$ for all $x\in\mathbb{R}^n$, then A must be the zero matrix.

(C) Suppose A and B are invertible square matrices. If A and B are similar to each other, then $A+A^T$ is similar to $B+B^T$.

(D) If A is an upper triangular (square) matrix and A is similar to a diagonal matrix, then A must be a diagonal matrix.

下列哪个论断是正确的? ( )

(A) 设 A 为 $m\times n$ 复矩阵, $B=A^HA$. (这里 $A^H=\overline{A}^T$ 表示 A 的共轭转置.) 则矩阵 $B+iI_n$ 可逆.

(B) 如果 A 是 n 阶实方阵, 且对任意 $x\in\mathbb{R}^n$ 均有 $x^TAx=0$, 则 A 一定是零矩阵.

(C) 假设 A 和 B 都是可逆方阵. 如果 A 和 B 相似, 则 $A+A^T$ 和 $B+B^T$ 相似.

(D) 若 A 是上三角(方)阵并且 A 相似于某个对角阵, 则 A 一定是对角阵.

Solution. (1) B (2) A (3) C (4) B (5) D.

2. (25 points, 5 points each) Fill in the blanks. (共 25 分, 每小题 5 分) 填空题.

(1) Let A be a 5×4 matrix of rank 3. Then the dimension of its left null space $N(A^T)$ is ＿＿＿＿＿

设 A 为秩等于 3 的 5×4 矩阵. 则它的左零空间 $N(A^T)$ 维数是 ＿＿＿＿＿

(2) Let A be a square matrix of order 3 and let $\alpha_1, \alpha_2, \alpha_3$ be linearly independent vectors in $\mathbb{R}^3$. Suppose $A\alpha_1=2\alpha_1+\alpha_2+\alpha_3$, $A\alpha_2=\alpha_2+2\alpha_3$ and $A\alpha_3=-\alpha_2+\alpha_3$. Then the real eigenvalues of A are ＿＿＿＿＿

设 A 为 3 阶方阵, $\alpha_1, \alpha_2, \alpha_3$ 为 $\mathbb{R}^3$ 中的线性无关向量组. 假设 $A\alpha_1=2\alpha_1+\alpha_2+\alpha_3$, $A\alpha_2=\alpha_2+2\alpha_3$ 且 $A\alpha_3=-\alpha_2+\alpha_3$. 则 A 的实特征值为 ＿＿＿＿＿

(3) Let A, B be square matrices of order n. Suppose $|A|=3$, $|B|=2$ and $|A^{-1}+B|=2$. Then $|A+B^{-1}|=$＿＿＿＿＿

设 A, B 为 n 阶方阵. 假设 $|A|=3$, $|B|=2$ 而 $|A^{-1}+B|=2$. 则 $|A+B^{-1}|=$＿＿＿＿＿

(4) Let ξ_1, ξ_2, ξ_3 be a basis of $\mathbb{R}^3$, and let $\eta_1 = \xi_1$, $\eta_2 = \xi_1 + \xi_2$, $\eta_3 = \xi_3 - \xi_1$. Let $\mathrm{Id} : \mathbb{R}^3 \to \mathbb{R}^3$ be the identity map.

Then the matrix A representing Id in the basis η_1, η_2, η_3 is ___________

设 ξ_1, ξ_2, ξ_3 为 $\mathbb{R}^3$ 的一组基, $\eta_1 = \xi_1$, $\eta_2 = \xi_1 + \xi_2$, $\eta_3 = \xi_3 - \xi_1$. 记 $\mathrm{Id} : \mathbb{R}^3 \to \mathbb{R}^3$ 为恒等映射. 则 Id 在 η_1, η_2, η_3 这组基下的矩阵表示是 $A =$___________

(5) Let $a \in \mathbb{R}$ and let $C \subseteq \mathbb{R}^2$ be the zero locus of the equation $3x^2 - 2axy + 3y^2 - 1 = 0$. Determine for which values of a the curve C is an ellipse. (A circle is also considered as an ellipse.) ___________

设 $a \in \mathbb{R}$, 记 $C \subseteq \mathbb{R}^2$ 为方程 $3x^2 - 2axy + 3y^2 - 1 = 0$ 的零点集. 求 a 的取值范围使得曲线 C 是个椭圆. (注意圆也认为是椭圆.) ___________

Solution. (1) 2 (2) 1, 2, 2 (3) 3 (4) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (5) $-3 < a < 3$.

3. (12 points)

(a) Find an orthonormal basis for the column space of

$$A = \begin{bmatrix} 1 & -2 & -1 \\ 2 & 0 & 1 \\ 2 & -4 & 2 \\ 4 & 0 & 0 \end{bmatrix}.$$

(b) Write A as QR, where Q has orthonormal columns and R is upper triangular.

(12 分)

(a) 找出下列矩阵的列空间的一组标准正交 (也称规范正交) 基:

$$A = \begin{bmatrix} 1 & -2 & -1 \\ 2 & 0 & 1 \\ 2 & -4 & 2 \\ 4 & 0 & 0 \end{bmatrix}.$$

(b) 将 A 分解成乘积 QR 的形式, 其中 Q 的列是标准正交 (也称规范正交) 的向量, R 是上三角矩阵.

Solution.

(a)

$$\begin{bmatrix} \frac{1}{5} \\ \frac{2}{5} \\ \frac{2}{5} \\ \frac{4}{5} \end{bmatrix}, \begin{bmatrix} -\frac{2}{5} \\ \frac{1}{5} \\ -\frac{4}{5} \\ \frac{2}{5} \end{bmatrix}, \begin{bmatrix} -\frac{4}{5} \\ \frac{2}{5} \\ \frac{2}{5} \\ -\frac{1}{5} \end{bmatrix}.$$

(b)

$$Q=\begin{bmatrix}\frac{1}{5} & -\frac{2}{5} & -\frac{4}{5}\\ \frac{2}{5} & \frac{1}{5} & \frac{2}{5}\\ \frac{2}{5} & -\frac{4}{5} & \frac{2}{5}\\ \frac{4}{5} & \frac{2}{5} & -\frac{1}{5}\end{bmatrix},\ R=\begin{bmatrix}5 & -2 & 1\\ 2 & 0 & 1\\ 0 & 0 & 2\end{bmatrix}.$$

4. (10 points) Let x, $y\in\mathbb{R}$, $n\geq 2$ and consider the following determinant of order n:

$$D_n(x,\,y)=\begin{vmatrix}x+y & xy & 0 & \dots & 0 & 0\\ 1 & x+y & xy & \dots & 0 & 0\\ 0 & 1 & x+y & \dots & 0 & 0\\ \dots & \dots & \dots & \dots & \dots & \dots\\ 0 & 0 & 0 & \dots & 1 & x+y\end{vmatrix}$$

For $n=2$, we consider $D_2(x,\,y)$ as $\begin{vmatrix}x+y & xy\\ 1 & x+y\end{vmatrix}$.

(a) Find a recurrence relation relating $D_n(x,\,y)$ to $D_{n-1}(x,\,y)$ and $D_{n-2}(x,\,y)$ for $n\geq 4$.

(b) Compute the determinant $D_n(x,\,y)$ for all $n\geq 2$.

(10 分) 设 x, $y\in\mathbb{R}$, $n\geq 2$. 考虑以下 n 阶行列式:

$$D_n(x,\,y)=\begin{vmatrix}x+y & xy & 0 & \dots & 0 & 0\\ 1 & x+y & xy & \dots & 0 & 0\\ 0 & 1 & x+y & \dots & 0 & 0\\ \dots & \dots & \dots & \dots & \dots & \dots\\ 0 & 0 & 0 & \dots & 1 & x+y\end{vmatrix}$$

当 $n=2$ 时, 我们认为 $D_2(x,\,y)$ 是 $\begin{vmatrix}x+y & xy\\ 1 & x+y\end{vmatrix}$.

(a) 在 $n\geq 4$ 时找出能将 $D_n(x,\,y)$ 和 $D_{n-1}(x,\,y), D_{n-2}(x,\,y)$ 建立联系的一个递推关系式.

(b) 对任意 $n\geq 2$ 计算行列式 $D_n(x,\,y)$.

Solution.

(a)

$$D_n=(x+y)D_{n-1}-xyD_{n-2}.$$

(b) $D_n(x,\,y)=x^n+x^{n-1}y+x^{n-2}y^2+\cdots+x^2y^{n-2}+xy^{n-1}+y^n$.

5. (12 points) Let a, $b\in\mathbb{R}$ and put

$$A=\begin{bmatrix}2 & -1 & 2\\ 5 & a & 3\\ -1 & b & -2\end{bmatrix}.$$

Suppose that $p=(1,1,-1)^T$ is an eigenvector of A.

(a) Find the values of a, b and find the eigenvalue λ corresponding to the eigenvector p.

(b) Is the matrix A diagonalizable? Please explain your answer.

(12 分) 设 $a, b \in \mathbb{R}$, 令

$$A = \begin{bmatrix} 2 & -1 & 2 \\ 5 & a & 3 \\ -1 & b & -2 \end{bmatrix}.$$

假设 $p = (1, 1, -1)^T$ 是 A 的一个特征向量.

(a) 求 a, b 的值并找出特征向量 p 对应的特征值 λ.

(b) 矩阵 A 是否可对角化？请解释你的答案.

Solution.

(a) $a = -3$, $b = 0$, $\lambda = -1$.

(b) The matrix is not diagonalizable, $\lambda = -1$ has multiplicity 3, but only has an eigenspace of dimension 1.

6. (12 points) Consider the ternary quadratic form

$$f(x_1, x_2, x_3) = (x_1 - x_2 + x_3)^2 + (x_2 + x_3)^2 + (x_1 + ax_3)^2$$

where $a \in \mathbb{R}$ is a parameter.

(a) What are the possible values of a if the quadratic form f is positive definite?

(b) What are the possible values of a if the equation $f(x_1, x_2, x_3) = 0$ has infinitely many solutions?

(c) Let $y = (y_1, y_2, y_3)^T$ be a new system of variables. Suppose that $f(x_1, x_2, x_3) = 0$ has infinitely many solutions.
Find an invertible linear transformation $y = Px$, where $x = (x_1, x_2, x_3)^T$, such that in the variables y_1, y_2, y_3 the quadratic form f has a diagonal form.

(12 分) 考虑三元二次型

$$f(x_1, x_2, x_3) = (x_1 - x_2 + x_3)^2 + (x_2 + x_3)^2 + (x_1 + ax_3)^2$$

其中 $a \in \mathbb{R}$ 为参数.

(a) 如果 f 是正定的, a 的取值范围是什么？

(b) 如果方程 $f(x_1, x_2, x_3) = 0$ 有无穷多个解, a 的取值可能是什么？

(c) 设 $y = (y_1, y_2, y_3)^T$ 为一组新的变元. 假设 $f(x_1, x_2, x_3) = 0$ 有无穷多个解.
找出一个可逆线性变换 $y = Px$, 其中 $x = (x_1, x_2, x_3)^T$, 使得在 y_1, y_2, y_3 这组变元下二次型 f 可以表示为对角形式.

Solution.

(a) $a \neq -2$.

(b) $a = 2$.

(c)

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & 2 \\ 1 & 0 & 2 \end{bmatrix}$$

7. (8 points) Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & -1 \end{bmatrix}$.

(a) Find all the singular values of A.

(b) Find the singular value decomposition of A, in other words, find two orthogonal matrices U and V (of suitable size) such that $A = U\Sigma V^T$.

(8 分) 令 $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & -1 \end{bmatrix}$.

(a) 求 A 的所有奇异值.

(b) 求 A 的奇异值分解. 即, 找出两个 (适当大小的) 正交矩阵 U 和 V 使得 $A = U\Sigma V^T$.

Solution.

(a) The singular values of A are 2, $\sqrt{2}$.

(b)

$$U = \begin{bmatrix} -\frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \end{bmatrix}, V = \begin{bmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix}, \Sigma = \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix}.$$

8. (6 points) Find a real 4×4 orthogonal matrix A such that A has no real eigenvalues but both A^2 and A^3 have real eigenvalues. Please explain why the matrix you give has the required properties.

(6 分) 给出一个 4×4 实正交矩阵 A 使得 A 没有实特征值, 而 A^2 和 A^3 都有实特征值. 请解释为什么你给出的矩阵满足要求.

Solution.

$$A = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

has no real eigenvalues.

However,

$$A^2 = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 2 & 0 \end{bmatrix} \text{ and } A^3 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & -2 & 2 \end{bmatrix}$$

have real eigenvalues.

Solutions

南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

考试科目: 线性代数 A　　开课单位: 数学系
考试时长: 120 分钟　　命题教师: 线性代数教学团队

题号	1	2	3	4	5	6	7	8
分值	15 分	25 分	10 分	16 分	10 分	6 分	16 分	12 分

本试卷共 (8) 大题, 满分 (110) 分. 请将所有答案写在答题本上.

This exam includes 8 questions and the score is 110 in total. Write all your answers on the examination book.

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1. (15 points, 3 points each) Multiple Choice. Only one choice is correct.

(共 15 分, 每小题 3 分) 选择题, 只有一个选项是正确的.

(1) Let A be an $m \times n$ real matrix and b be a column vector in $\mathbb{R}^m$. Which of the following statements is correct ?

A

(A) If $Ax = b$ has infinite many solutions, then $Ax = 0$ has a nonzero solution.

(B) If the system $Ax = 0$ has only zero solution, then $Ax = b$ has one and only one solution.

(C) If the rank of A is n, then the system $Ax = b$ must have a solution.

(D) If A is a square matrix (i.e., $m = n$), then the system $Ax = b$ is consistent if and only if A is invertible.

设 A 为 $m \times n$ 实矩阵, b 是 $\mathbb{R}^m$ 中的列向量. 下列陈述中哪个是正确的?

(A) 如果 $Ax = b$ 有无穷多个解, 则 $Ax = 0$ 有非零解.

(B) 如果方程组 $Ax = 0$ 只有零解, 则 $Ax = b$ 有且仅有一个解.

(C) 如果 A 的秩为 n, 则方程组 $Ax = b$ 必有解.

(D) 如果 A 是方阵 (即 $m = n$), 则方程组 $Ax = b$ 是相容的当且仅当 A 可逆.

C

(2) Suppose A is an $m \times n$ matrix, B is an $n \times m$ matrix, and I is the $m \times m$ identity matrix. If $AB = I$, then ( )

(A) the column vectors of A are linearly independent, and the row vectors of B are linearly independent.

(B) the column vectors of A are linearly independent, and the column vectors of B are linearly independent.

(C) the row vectors of A are linearly independent, and the column vectors of B are linearly independent.

(D) the row vectors of A are linearly independent, and the row vectors of B are linearly independent.

设 A 为 $m\times n$ 型矩阵, B 为 $n\times m$ 型矩阵, I 为 m 阶单位矩阵. 若 $AB=I$, 则 (　　)

(A) A 的列向量组线性无关, B 的行向量组线性无关.

(B) A 的列向量组线性无关, B 的列向量组线性无关.

(C) A 的行向量组线性无关, B 的列向量组线性无关.

(D) A 的行向量组线性无关, B 的行向量组线性无关.

D

(3) Let A be a 3×3 matrix, and let B be the matrix formed by adding the second column of A to its first column. Suppose that after exchanging the second and third rows of B, the resulting matrix is the 3×3 identity matrix. Let $P_1=\begin{bmatrix}1&0&0\\1&1&0\\0&0&1\end{bmatrix}$, $P_2=\begin{bmatrix}1&0&0\\0&0&1\\0&1&0\end{bmatrix}$. Then $A=$ (　　)

(A) P_1P_2.

(B) $P_1^{-1}P_2$.

(C) P_2P_1.

(D) $P_2P_1^{-1}$.

设 A 为 3 阶方阵, 将 A 的第二列加到第一列得矩阵 B. 假设交换 B 的第二行与第三行可以得到 3 阶单位矩阵. 记 $P_1=\begin{bmatrix}1&0&0\\1&1&0\\0&0&1\end{bmatrix}$, $P_2=\begin{bmatrix}1&0&0\\0&0&1\\0&1&0\end{bmatrix}$. 则 $A=$ (　　)

(A) P_1P_2.

(B) $P_1^{-1}P_2$.

(C) P_2P_1.

(D) $P_2P_1^{-1}$.

D

(4) Let $A=\begin{bmatrix}-1&2&3\\-3&6&8\\2&-4&t\end{bmatrix}$, where $t\in\mathbb{R}$. Suppose $\operatorname{rank}(A)=2$. Then (　　)

(A) $t=-6$.

(B) $t=6$.

(C) $t\neq 0$.

(D) t can be any real number.

设 $A=\begin{bmatrix}-1&2&3\\-3&6&8\\2&-4&t\end{bmatrix}$, 其中 $t\in\mathbb{R}$. 假设 $\operatorname{rank}(A)=2$. 则 (　　)

(A) $t=-6$.

(B) $t=6$.

(C) $t \neq 0$.

(D) t 可以是任意实数.

C

(5) Which of the following statements is incorrect? (　　)

(A) For any matrix A, $\operatorname{rank}(A) = \dim C(A)$.

(B) If $v_1, \cdots, v_m$ are pairwise orthogonal nonzero vectors, then the vectors $v_1, \cdots, v_m$ are linear independent.

(C) If A is an upper triangular $n \times n$ matrix such that $A^2 = 0$, then $A = 0$.

(D) Let A, B be $n \times n$ matrices such that AB is invertible. Then both A and B are invertible.

下列哪个论断是错误的? (　　)

(A) 对于任意矩阵 A, $\operatorname{rank}(A) = \dim C(A)$.

(B) 如果 $v_1, \cdots, v_m$ 是一组两两正交的非零向量, 则向量组 $v_1, \cdots, v_m$ 线性无关.

(C) 如果 A 是 $n \times n$ 上三角矩阵且 $A^2 = 0$, 则 $A = 0$.

(D) 设 A, B 为 $n \times n$ 矩阵且 AB 可逆. 则 A 和 B 都可逆.

2. (25 points, 5 points each) Fill in the blanks. (共 25 分, 每小题 5 分) 填空题.

(1) Let A, B be invertible $n \times n$ matrices. Then the inverse of the block matrix $\begin{bmatrix} 0 & A \\ B & 0 \end{bmatrix}$ is ________

设 A, B 均为 $n \times n$ 可逆矩阵. 则分块矩阵 $\begin{bmatrix} 0 & A \\ B & 0 \end{bmatrix}$ 的逆为 $\begin{bmatrix} 0 & B^{-1} \\ A^{-1} & 0 \end{bmatrix}$

(2) Suppose A is a 3×4 matrix and $\dim N(A) = 2$. Then $\dim N(A^T) =$ ________

设 A 为 3×4 矩阵且 $\dim N(A) = 2$. 则 $\dim N(A^T) =$ 1

(3) Let $A = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix}$. Then $A^{-1} =$ $\begin{bmatrix} 1 & 0 & 0 \\ -a & 1 & 0 \\ ac-b & -c & 1 \end{bmatrix}$

设 $A = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix}$. 则 $A^{-1} =$ ________

(4) Let u, v be vectors in $\mathbb{R}^n$ such that $\|u\| = 3$, $\|v\| = 4$ and $u^T v = -3$.

Then $\|2u + 3v\| =$ ________

设 u, v 为 $\mathbb{R}^n$ 中的向量, 满足 $\|u\| = 3$, $\|v\| = 4$ 以及 $u^T v = -3$. 则 $\|2u+3v\| =$ 12

(5) Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & -1 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}$.

Then the least squares solution to $Ax = b$ is $\hat{x} =$ $\begin{bmatrix} 4 \\ -5 \end{bmatrix}$

设 $A=\begin{bmatrix}1&1\\1&0\\0&-1\end{bmatrix}$, $b=\begin{bmatrix}1\\2\\7\end{bmatrix}$.

则 $Ax=b$ 的最小二乘解是 $\hat{x}=$__________

3. (10 points) Find the LU factorization of the matrix $A=\begin{bmatrix}3&1&1\\1&3&1\\1&1&3\end{bmatrix}$.

求矩阵 $A=\begin{bmatrix}3&1&1\\1&3&1\\1&1&3\end{bmatrix}$ 的 LU 分解.

$L=\begin{bmatrix}1&0&0\\\frac{1}{3}&1&0\\\frac{1}{3}&\frac{1}{4}&1\end{bmatrix}$ $U=\begin{bmatrix}3&1&1\\0&\frac{8}{3}&\frac{2}{3}\\0&0&\frac{5}{2}\end{bmatrix}$

4. (16 points) Let $A=\begin{bmatrix}0&1&2&3&4\\0&1&2&4&6\\0&0&0&1&2\end{bmatrix}$.

Please give a basis for each of the four fundamental subspaces $C(A)$, $N(A)$, $C(A^T)$ and $N(A^T)$, respectively.

$C(A)=\text{Span}\left\{\begin{bmatrix}1\\1\\0\end{bmatrix},\begin{bmatrix}0\\1\\1\end{bmatrix}\right\}$

(16 分) 设 $A=\begin{bmatrix}0&1&2&3&4\\0&1&2&4&6\\0&0&0&1&2\end{bmatrix}$.

$N(A)=\text{Span}\left\{\begin{bmatrix}1\\0\\0\\0\\0\end{bmatrix},\begin{bmatrix}0\\2\\-1\\0\\0\end{bmatrix},\begin{bmatrix}0\\-2\\0\\2\\-1\end{bmatrix}\right\}$

请对四个基本子空间 $C(A)$, $N(A)$, $C(A^T)$ 和 $N(A^T)$ 分别给出各自的一组基.

$C(A^T)=\text{Span}\left\{\begin{bmatrix}0\\1\\2\\3\\4\end{bmatrix},\begin{bmatrix}0\\0\\0\\1\\2\end{bmatrix}\right\}$

$N(A^T)=\text{Span}\left\{\begin{bmatrix}1\\-1\\1\end{bmatrix}\right\}$

5. (10 points) Let $E=\{u_1,u_2,u_3\}$ and $F=\{v_1,v_2\}$, where

$$u_1=\begin{bmatrix}1\\0\\-1\end{bmatrix},\ u_2=\begin{bmatrix}1\\2\\1\end{bmatrix},\ u_3=\begin{bmatrix}-1\\1\\1\end{bmatrix}\quad\text{and}\quad v_1=\begin{bmatrix}1\\-1\end{bmatrix},\ v_2=\begin{bmatrix}2\\-1\end{bmatrix}.$$

Define the linear transformation $T:\mathbb{R}^3\to\mathbb{R}^2$ by

$$T\left(\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}\right)=\begin{bmatrix}2x_2\\-x_1\end{bmatrix}.$$

Find the matrix A representing T with respect to the ordered bases E and F.

$A=\begin{bmatrix}2&-2&-4\\-1&3&3\end{bmatrix}$

(10 分) 设 $E=\{u_1,u_2,u_3\}$, $F=\{v_1,v_2\}$, 其中

$$u_1=\begin{bmatrix}1\\0\\-1\end{bmatrix},\ u_2=\begin{bmatrix}1\\2\\1\end{bmatrix},\ u_3=\begin{bmatrix}-1\\1\\1\end{bmatrix}\quad\text{and}\quad v_1=\begin{bmatrix}1\\-1\end{bmatrix},\ v_2=\begin{bmatrix}2\\-1\end{bmatrix}.$$

定义线性变换 $T: \mathbb{R}^3 \to \mathbb{R}^2$ 如下

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} 2x_2 \\ -x_1 \end{bmatrix}.$$

求 T 在 E 和 F 这两组有序基下的矩阵表示 A.

6. (6 points) Let A, B be $n\times n$ matrices. Suppose A and B are both symmetric. Is AB necessarily symmetric? If yes, please give a proof. Otherwise please give a counterexample.

(6 分) 设 A, B 均为 $n \times n$ 矩阵. 假设 A 和 B 都是对称矩阵. AB 是否一定是对称矩阵? 若是, 请给出证明. 否则请给出一个反例. $A=\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ $B=\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ $AB=\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$

7. (16 points) The following two questions are independent:

(a) Let A be the 2×2 matrix such that the linear transformation $\mathbb{R}^2 \to \mathbb{R}^2$, $v \mapsto Av$ rotate every vector in $\mathbb{R}^2$ through $60°$ counter-clockwise (about the origin). Find A and A^{2020}.

$$A=\begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \cos\frac{\pi}{3} & -\sin\frac{\pi}{3} \\ \sin\frac{\pi}{3} & \cos\frac{\pi}{3} \end{pmatrix}$$

(b) Three planes Π_1, Π_2, Π_3 in the space $\mathbb{R}^3$ are given by the equations

$$\Pi_1 : \quad x+y+z=0,$$
$$\Pi_2 : \quad 2x-y+4z=0,$$
$$\Pi_3 : \quad -x+2y-z=0.$$

Determine a matrix representative (in the standard basis of $\mathbb{R}^3$) of a linear transformation taking the xy plane to Π_1, the yz plane to Π_2 and the zx plane to Π_3.

$$A^{2020} = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} \cos\frac{4\pi}{3} & -\sin\frac{4\pi}{3} \\ \sin\frac{4\pi}{3} & \cos\frac{4\pi}{3} \end{pmatrix}$$

(16 分) 以下两个小题是相互独立的:

(a) 设 A 是 2×2 矩阵使得线性变换 $\mathbb{R}^2 \to \mathbb{R}^2$, $v \mapsto Av$ 把 $\mathbb{R}^2$ 中每个向量 (绕原点) 逆时针转动 $60°$.
求 A 和 A^{2020}.

(b) 在空间 $\mathbb{R}^3$ 中由以下方程给出三个平面 Π_1, Π_2, Π_3:

$$\Pi_1 : \quad x+y+z=0,$$
$$\Pi_2 : \quad 2x-y+4z=0,$$
$$\Pi_3 : \quad -x+2y-z=0.$$

求一个矩阵, 使它 (在 $\mathbb{R}^3$ 的标准基下) 表示的线性变换将 xy 平面映射成 Π_1, 将 yz 平面映射成 Π_2 并将 zx 平面映射成 Π_3.

例如 $\begin{pmatrix} -1 & -5 & -7 \\ 0 & 2 & -2 \\ 1 & 3 & 3 \end{pmatrix}$

每一列可替换为它的任意非零常数倍

8. (12 points) Let A be a 3×3 matrix such that $\mathrm{rank}(A)=2$ and $A^3=0$.

(a) Prove that $\mathrm{rank}(A^2)=1$.

(b) Let $\alpha_1\in\mathbb{R}^3$ be a nonzero vector such that $A\alpha_1=0$. Prove that there exist vectors α_2, α_3 such that $A\alpha_2=\alpha_1$, $A^2\alpha_3=\alpha_1$.

(c) For any vectors α_2,α_3 described as above, show that $\alpha_1,\alpha_2,\alpha_3$ are linearly independent.

(In this problem, you are allowed to assume the statements of some questions to answer subsequent questions.)

(12 分) 设 A 是 3×3 矩阵, 它满足 $\mathrm{rank}(A)=2$ 及 $A^3=0$.

(a) 证明 $\mathrm{rank}(A^2)=1$.

(b) 设 $\alpha_1\in\mathbb{R}^3$ 是满足 $A\alpha_1=0$ 的非零向量. 证明: 存在向量 α_2, α_3 使得 $A\alpha_2=\alpha_1$, $A^2\alpha_3=\alpha_1$.

(c) 证明: 对于任意满足上述条件的向量 α_2,α_3, 向量组 $\alpha_1,\alpha_2,\alpha_3$ 线性无关.

(本题中, 允许承认前面小题的结果来用于后续问题的解答.)

(a)
$$\left.\begin{array}{l} A^3=0 \Rightarrow C(A^2)\subseteq N(A) \\ \mathrm{rank}(A)=2 \Rightarrow \dim N(A)=1 \end{array}\right\} \Rightarrow \dim C(A^2)=\mathrm{rank}(A^2)\le 1$$

If $A^2=0$ then $C(A)\subseteq N(A)$. Then $\dim C(A)=\mathrm{rank}(A)\le \dim N(A)=1$

contradiction to $\mathrm{rank}(A)=2$

So $0<\mathrm{rank}(A^2)\le 1$, $\mathrm{rank}(A^2)=1$.

(b) We have $C(A^2)\subseteq N(A)$ and $N(A)=\mathrm{Span}(\alpha_1)$.

By (a) + Dimension count $\Rightarrow$ $C(A^2)=\mathrm{span}(\alpha_1)$

$\Rightarrow \exists\, \alpha_3$ s.t. $A^2\alpha_3=\alpha_1$

Then take $\alpha_2=A\alpha_3$.

(c) $x_1\alpha_1+x_2\alpha_2+x_3\alpha_3=0$

$\Rightarrow A^2(x_1\alpha_1+x_2\alpha_2+x_3\alpha_3)=0 \Rightarrow x_1\alpha_3=0$

$\Rightarrow x_1=0$

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本试卷共 (8) 大题, 满分 (110) 分. 请将所有答案写在答题本上.

This exam includes 8 questions and the score is 110 in total. Write all your answers on the examination book.

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1. (15 points, 3 points each) Multiple Choice. Only one choice is correct.

   (共 15 分, 每小题 3 分) 选择题, 只有一个选项是正确的.

   (1) Let A be an $m \times n$ real matrix and b be a column vector in $\mathbb{R}^m$. Which of the following statements is correct? (　　)

   (A) If $Ax = b$ has infinite many solutions, then $Ax = 0$ has a nonzero solution.

   (B) If the system $Ax = 0$ has only zero solution, then $Ax = b$ has one and only one solution.

   (C) If the rank of A is n, then the system $Ax = b$ must have a solution.

   (D) If A is a square matrix (i.e., $m = n$), then the system $Ax = b$ is consistent if and only if A is invertible.

   设 A 为 $m \times n$ 实矩阵, b 是 $\mathbb{R}^m$ 中的列向量. 下列陈述中哪个是正确的? (　　)

   (A) 如果 $Ax = b$ 有无穷多个解, 则 $Ax = 0$ 有非零解.

   (B) 如果方程组 $Ax = 0$ 只有零解, 则 $Ax = b$ 有且仅有一个解.

   (C) 如果 A 的秩为 n, 则方程组 $Ax = b$ 必有解.

   (D) 如果 A 是方阵 (即 $m = n$), 则方程组 $Ax = b$ 是相容的当且仅当 A 可逆.

   (2) Suppose A is an $m \times n$ matrix, B is an $n \times m$ matrix, and I is the $m \times m$ identity matrix. If $AB = I$, then (　　)

   (A) the column vectors of A are linearly independent, and the row vectors of B are linearly independent.

   (B) the column vectors of A are linearly independent, and the column vectors of B are linearly independent.

   (C) the row vectors of A are linearly independent, and the column vectors of B are linearly independent.

(D) the row vectors of A are linearly independent, and the row vectors of B are linearly independent.

设 A 为 $m\times n$ 型矩阵, B 为 $n\times m$ 型矩阵, I 为 m 阶单位矩阵. 若 $AB=I$, 则 (　　)

(A) A 的列向量组线性无关, B 的行向量组线性无关.

(B) A 的列向量组线性无关, B 的列向量组线性无关.

(C) A 的行向量组线性无关, B 的列向量组线性无关.

(D) A 的行向量组线性无关, B 的行向量组线性无关.

(3) Let A be a 3×3 matrix, and let B be the matrix formed by adding the second column of A to its first column. Suppose that after exchanging the second and third rows of B, the resulting matrix is the 3×3 identity matrix. Let $P_1=\begin{bmatrix}1&0&0\\1&1&0\\0&0&1\end{bmatrix}$, $P_2=\begin{bmatrix}1&0&0\\0&0&1\\0&1&0\end{bmatrix}$. Then $A=$ (　　)

(A) P_1P_2.

(B) $P_1^{-1}P_2$.

(C) P_2P_1.

(D) $P_2P_1^{-1}$.

设 A 为 3 阶方阵, 将 A 的第二列加到第一列得矩阵 B. 假设交换 B 的第二行与第三行可以得到 3 阶单位矩阵. 记 $P_1=\begin{bmatrix}1&0&0\\1&1&0\\0&0&1\end{bmatrix}$, $P_2=\begin{bmatrix}1&0&0\\0&0&1\\0&1&0\end{bmatrix}$. 则 $A=$ (　　)

(A) P_1P_2.

(B) $P_1^{-1}P_2$.

(C) P_2P_1.

(D) $P_2P_1^{-1}$.

(4) Let $A=\begin{bmatrix}-1&2&3\\-3&6&8\\2&-4&t\end{bmatrix}$, where $t\in\mathbb{R}$. Suppose $\operatorname{rank}(A)=2$. Then (　　)

(A) $t=-6$.

(B) $t=6$.

(C) $t\neq 0$.

(D) t can be any real number.

设 $A=\begin{bmatrix}-1&2&3\\-3&6&8\\2&-4&t\end{bmatrix}$, 其中 $t\in\mathbb{R}$. 假设 $\operatorname{rank}(A)=2$. 则 (　　)

(A) $t=-6$.

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(5) Which of the following statements is incorrect? (　　)

(A) For any matrix A, $\operatorname{rank}(A) = \dim C(A)$.

(B) If $v_1, \cdots, v_m$ are pairwise orthogonal nonzero vectors, then the vectors $v_1, \cdots, v_m$ are linear independent.

(C) If A is an upper triangular $n \times n$ matrix such that $A^2 = 0$, then $A = 0$.

(D) Let A, B be $n \times n$ matrices such that AB is invertible. Then both A and B are invertible.

下列哪个论断是错误的? (　　)

(A) 对于任意矩阵 A, $\operatorname{rank}(A) = \dim C(A)$.

(B) 如果 $v_1, \cdots, v_m$ 是一组两两正交的非零向量, 则向量组 $v_1, \cdots, v_m$ 线性无关.

(C) 如果 A 是 $n \times n$ 上三角矩阵且 $A^2 = 0$, 则 $A = 0$.

(D) 设 A, B 为 $n \times n$ 矩阵且 AB 可逆. 则 A 和 B 都可逆.

2. (25 points, 5 points each) Fill in the blanks. (共 25 分, 每小题 5 分) 填空题.

(1) Let A, B be invertible $n \times n$ matrices. Then the inverse of the block matrix $\begin{bmatrix} 0 & A \\ B & 0 \end{bmatrix}$ is __________

设 A, B 均为 $n \times n$ 可逆矩阵. 则分块矩阵 $\begin{bmatrix} 0 & A \\ B & 0 \end{bmatrix}$ 的逆为 __________

(2) Suppose A is a 3×4 matrix and $\dim N(A) = 2$. Then $\dim N(A^T) =$ __________

设 A 为 3×4 矩阵且 $\dim N(A) = 2$. 则 $\dim N(A^T) =$ __________

(3) Let $A = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix}$. Then $A^{-1} =$ __________

设 $A = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix}$. 则 $A^{-1} =$ __________

(4) Let u, v be vectors in $\mathbb{R}^n$ such that $\|u\| = 3$, $\|v\| = 4$ and $u^T v = -3$.

Then $\|2u + 3v\| =$ __________

设 u, v 为 $\mathbb{R}^n$ 中的向量, 满足 $\|u\| = 3$, $\|v\| = 4$ 以及 $u^T v = -3$. 则 $\|2u+3v\| =$ __________

(5) Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & -1 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}$.

Then the least squares solution to $Ax = b$ is $\hat{x} =$ __________

设 $A=\begin{bmatrix}1&1\\1&0\\0&-1\end{bmatrix}$, $b=\begin{bmatrix}1\\2\\7\end{bmatrix}$.

则 $Ax=b$ 的最小二乘解是 $\hat{x}=$__________

3. (10 points) Find the LU factorization of the matrix $A=\begin{bmatrix}3&1&1\\1&3&1\\1&1&3\end{bmatrix}$.

求矩阵 $A=\begin{bmatrix}3&1&1\\1&3&1\\1&1&3\end{bmatrix}$ 的 LU 分解.

4. (16 points) Let $A=\begin{bmatrix}0&1&2&3&4\\0&1&2&4&6\\0&0&0&1&2\end{bmatrix}$.

Please give a basis for each of the four fundamental subspaces $C(A)$, $N(A)$, $C(A^T)$ and $N(A^T)$, respectively.

(16 分) 设 $A=\begin{bmatrix}0&1&2&3&4\\0&1&2&4&6\\0&0&0&1&2\end{bmatrix}$.

请对四个基本子空间 $C(A)$, $N(A)$, $C(A^T)$ 和 $N(A^T)$ 分别给出各自的一组基.

5. (10 points) Let $E=\{u_1, u_2, u_3\}$ and $F=\{v_1, v_2\}$, where

$$u_1=\begin{bmatrix}1\\0\\-1\end{bmatrix},\ u_2=\begin{bmatrix}1\\2\\1\end{bmatrix},\ u_3=\begin{bmatrix}-1\\1\\1\end{bmatrix}\quad\text{and}\quad v_1=\begin{bmatrix}1\\-1\end{bmatrix},\ v_2=\begin{bmatrix}2\\-1\end{bmatrix}.$$

Define the linear transformation $T:\mathbb{R}^3\to\mathbb{R}^2$ by

$$T\left(\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}\right)=\begin{bmatrix}2x_2\\-x_1\end{bmatrix}.$$

Find the matrix A representing T with respect to the ordered bases E and F.

(10 分) 设 $E=\{u_1, u_2, u_3\}$, $F=\{v_1, v_2\}$, 其中

$$u_1=\begin{bmatrix}1\\0\\-1\end{bmatrix},\ u_2=\begin{bmatrix}1\\2\\1\end{bmatrix},\ u_3=\begin{bmatrix}-1\\1\\1\end{bmatrix}\quad\text{and}\quad v_1=\begin{bmatrix}1\\-1\end{bmatrix},\ v_2=\begin{bmatrix}2\\-1\end{bmatrix}.$$

定义线性变换 $T:\mathbb{R}^3\to\mathbb{R}^2$ 如下

$$T\left(\begin{bmatrix} x_1\\ x_2\\ x_3\end{bmatrix}\right)=\begin{bmatrix} 2x_2\\ -x_1\end{bmatrix}.$$

求 T 在 E 和 F 这两组有序基下的矩阵表示 A.

6. (6 points) Let A, B be $n\times n$ matrices. Suppose A and B are both symmetric. Is AB necessarily symmetric? If yes, please give a proof. Otherwise please give a counterexample.

(6 分) 设 A, B 均为 $n\times n$ 矩阵. 假设 A 和 B 都是对称矩阵. AB 是否一定是对称矩阵？若是, 请给出证明. 否则请给出一个反例.

7. (16 points) The following two questions are independent:

   (a) Let A be the 2×2 matrix such that the linear transformation $\mathbb{R}^2\to\mathbb{R}^2$, $v\mapsto Av$ rotates every vector in $\mathbb{R}^2$ through 60° counter-clockwise (about the origin).
   Find A and A^{2020}.

   (b) Three planes Π_1, Π_2, Π_3 in the space $\mathbb{R}^3$ are given by the equations

$$\begin{aligned}\Pi_1 &:\ x+y+z=0\,,\\ \Pi_2 &:\ 2x-y+4z=0\,,\\ \Pi_3 &:\ -x+2y-z=0\,.\end{aligned}$$

   Determine a matrix representative (in the standard basis of $\mathbb{R}^3$) of a linear transformation taking the xy plane to Π_1, the yz plane to Π_2 and the zx plane to Π_3.

(16 分) 以下两个小题是相互独立的:

   (a) 设 A 是 2×2 矩阵使得线性变换 $\mathbb{R}^2\to\mathbb{R}^2$, $v\mapsto Av$ 把 $\mathbb{R}^2$ 中每个向量 (绕原点) 逆时针转动 60°.
   求 A 和 A^{2020}.

   (b) 在空间 $\mathbb{R}^3$ 中由以下方程给出三个平面 Π_1, Π_2, Π_3:

$$\begin{aligned}\Pi_1 &:\ x+y+z=0\,,\\ \Pi_2 &:\ 2x-y+4z=0\,,\\ \Pi_3 &:\ -x+2y-z=0\,.\end{aligned}$$

   求一个矩阵, 使它 (在 $\mathbb{R}^3$ 的标准基下) 表示的线性变换将 xy 平面映射成 Π_1, 将 yz 平面映射成 Π_2 并将 zx 平面映射成 Π_3.

8. (12 points) Let A be a 3×3 matrix such that $\operatorname{rank}(A) = 2$ and $A^3 = 0$.

(a) Prove that $\operatorname{rank}(A^2) = 1$.

(b) Let $\alpha_1 \in \mathbb{R}^3$ be a nonzero vector such that $A\alpha_1 = 0$. Prove that there exist vectors α_2, α_3 such that $A\alpha_2 = \alpha_1$, $A^2\alpha_3 = \alpha_1$.

(c) For any vectors α_2, α_3 described as above, show that $\alpha_1, \alpha_2, \alpha_3$ are linearly independent.

(In this problem, you are allowed to assume the statements of some questions to answer subsequent questions.)

(12 分) 设 A 是 3×3 矩阵, 它满足 $\operatorname{rank}(A) = 2$ 及 $A^3 = 0$.

(a) 证明 $\operatorname{rank}(A^2) = 1$.

(b) 设 $\alpha_1 \in \mathbb{R}^3$ 是满足 $A\alpha_1 = 0$ 的非零向量. 证明: 存在向量 α_2, α_3 使得 $A\alpha_2 = \alpha_1$, $A^2\alpha_3 = \alpha_1$.

(c) 证明: 对于任意满足上述条件的向量 α_2, α_3, 向量组 $\alpha_1, \alpha_2, \alpha_3$ 线性无关.

(本题中, 允许承认前面小题的结果来用于后续问题的解答.)

Solutions

南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

考试科目: 线性代数A　　　　开课单位: 数学系
考试时长: 120 分钟　　　　命题教师: 线性代数教学团队

题号	1	2	3	4	5	6	7	8
分值	15 分	25 分	10 分	16 分	10 分	6 分	16 分	12 分

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This exam includes 8 questions and the score is 110 in total. Write all your answers on the examination book.

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1. (15 points, 3 points each) Multiple Choice. Only one choice is correct.

(共 15 分, 每小题 3 分) 选择题, 只有一个选项是正确的.

(1) Let A be an $m \times n$ real matrix and b be a column vector in $\mathbb{R}^m$. Which of the following statements is correct?

A

(A) If $Ax = b$ has infinite many solutions, then $Ax = 0$ has a nonzero solution.

(B) If the system $Ax = 0$ has only zero solution, then $Ax = b$ has one and only one solution.

(C) If the rank of A is n, then the system $Ax = b$ must have a solution.

(D) If A is a square matrix (i.e., $m = n$), then the system $Ax = b$ is consistent if and only if A is invertible.

设 A 为 $m \times n$ 实矩阵, b 是 $\mathbb{R}^m$ 中的列向量. 下列陈述中哪个是正确的？

(A) 如果 $Ax = b$ 有无穷多个解, 则 $Ax = 0$ 有非零解.

(B) 如果方程组 $Ax = 0$ 只有零解, 则 $Ax = b$ 有且仅有一个解.

(C) 如果 A 的秩为 n, 则方程组 $Ax = b$ 必有解.

(D) 如果 A 是方阵 (即 $m = n$), 则方程组 $Ax = b$ 是相容的当且仅当 A 可逆.

C

(2) Suppose A is an $m \times n$ matrix, B is an $n \times m$ matrix, and I is the $m \times m$ identity matrix. If $AB = I$, then (　　)

(A) the column vectors of A are linearly independent, and the row vectors of B are linearly independent.

(B) the column vectors of A are linearly independent, and the column vectors of B are linearly independent.

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设 A 为 $m \times n$ 型矩阵, B 为 $n \times m$ 型矩阵, I 为 m 阶单位矩阵. 若 $AB = I$, 则 (　　)

(A) A 的列向量组线性无关, B 的行向量组线性无关.

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(C) A 的行向量组线性无关, B 的列向量组线性无关.

(D) A 的行向量组线性无关, B 的行向量组线性无关.

D

(3) Let A be a 3×3 matrix, and let B be the matrix formed by adding the second column of A to its first column. Suppose that after exchanging the second and third rows of B, the resulting matrix is the 3×3 identity matrix. Let $P_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$. Then $A =$ (　　)

(A) $P_1 P_2$.

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设 A 为 3 阶方阵, 将 A 的第二列加到第一列得矩阵 B. 假设交换 B 的第二行与第三行可以得到 3 阶单位矩阵. 记 $P_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$. 则 $A =$ (　　)

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D

(4) Let $A = \begin{bmatrix} -1 & 2 & 3 \\ -3 & 6 & 8 \\ 2 & -4 & t \end{bmatrix}$, where $t \in \mathbb{R}$. Suppose $\operatorname{rank}(A) = 2$. Then (　　)

(A) $t = -6$.

(B) $t = 6$.

(C) $t \neq 0$.

(D) t can be any real number.

设 $A = \begin{bmatrix} -1 & 2 & 3 \\ -3 & 6 & 8 \\ 2 & -4 & t \end{bmatrix}$, 其中 $t \in \mathbb{R}$. 假设 $\operatorname{rank}(A) = 2$. 则 (　　)

(A) $t = -6$.

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C (5) Which of the following statements is incorrect ? ( )

(A) For any matrix A, $\operatorname{rank}(A) = \dim C(A)$.

(B) If $v_1, \cdots, v_m$ are pairwise orthogonal nonzero vectors, then the vectors $v_1, \cdots, v_m$ are linear independent.

(C) If A is an upper triangular $n \times n$ matrix such that $A^2 = 0$, then $A = 0$.

(D) Let A, B be $n \times n$ matrices such that AB is invertible. Then both A and B are invertible.

下列哪个论断是错误的？( )

(A) 对于任意矩阵 A, $\operatorname{rank}(A) = \dim C(A)$.

(B) 如果 $v_1, \cdots, v_m$ 是一组两两正交的非零向量, 则向量组 $v_1, \cdots, v_m$ 线性无关.

(C) 如果 A 是 $n \times n$ 上三角矩阵且 $A^2 = 0$, 则 $A = 0$.

(D) 设 A, B 为 $n \times n$ 矩阵且 AB 可逆. 则 A 和 B 都可逆.

2. (25 points, 5 points each) Fill in the blanks. (共 25 分, 每小题 5 分) 填空题.

(1) Let A, B be invertible $n \times n$ matrices. Then the inverse of the block matrix $\begin{bmatrix} 0 & A \\ B & 0 \end{bmatrix}$ is ________

设 A, B 均为 $n \times n$ 可逆矩阵. 则分块矩阵 $\begin{bmatrix} 0 & A \\ B & 0 \end{bmatrix}$ 的逆为 $\begin{bmatrix} 0 & B^{-1} \\ A^{-1} & 0 \end{bmatrix}$

(2) Suppose A is a 3×4 matrix and $\dim N(A) = 2$. Then $\dim N(A^T) =$ ________

设 A 为 3×4 矩阵且 $\dim N(A) = 2$. 则 $\dim N(A^T) =$ 1

(3) Let $A = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix}$. Then $A^{-1} =$ $\begin{bmatrix} 1 & 0 & 0 \\ -a & 1 & 0 \\ ac-b & -c & 1 \end{bmatrix}$

设 $A = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix}$. 则 $A^{-1} =$ ________

(4) Let u, v be vectors in $\mathbb{R}^n$ such that $\|u\| = 3$, $\|v\| = 4$ and $u^T v = -3$.

Then $\|2u + 3v\| =$ ________

设 u, v 为 $\mathbb{R}^n$ 中的向量, 满足 $\|u\| = 3$, $\|v\| = 4$ 以及 $u^T v = -3$. 则 $\|2u+3v\| =$ 12

(5) Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & -1 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}$.

Then the least squares solution to $Ax = b$ is $\hat{x} =$ $\begin{bmatrix} 4 \\ -5 \end{bmatrix}$

设 $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & -1 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}$.

则 $Ax = b$ 的最小二乘解是 $\hat{x} =$__________

3. (10 points) Find the LU factorization of the matrix $A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}$.

求矩阵 $A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}$ 的 LU 分解.

$L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & 1 & 0 \\ \frac{1}{3} & \frac{1}{4} & 1 \end{bmatrix}$ $U = \begin{bmatrix} 3 & 1 & 1 \\ 0 & \frac{8}{3} & \frac{2}{3} \\ 0 & 0 & \frac{5}{2} \end{bmatrix}$.

4. (16 points) Let $A = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 4 & 6 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$.

Please give a basis for each of the four fundamental subspaces $C(A)$, $N(A)$, $C(A^T)$ and $N(A^T)$, respectively.

(16 分) 设 $A = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 4 & 6 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$.

请对四个基本子空间 $C(A)$, $N(A)$, $C(A^T)$ 和 $N(A^T)$ 分别给出各自的一组基.

$C(A) = \text{Span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$

$N(A) = \text{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 0 \\ 2 \\ -1 \end{bmatrix} \right\}$

$C(A^T) = \text{Span}\left\{ \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 2 \end{bmatrix} \right\}$

$N(A^T) = \text{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$

5. (10 points) Let $E = \{u_1, u_2, u_3\}$ and $F = \{v_1, v_2\}$, where

$$u_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \ u_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \ u_3 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \quad \text{and} \quad v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \ v_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Define the linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$ by

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} 2x_2 \\ -x_1 \end{bmatrix}.$$

Find the matrix A representing T with respect to the ordered bases E and F.

$A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 3 \end{bmatrix}$

(10 分) 设 $E = \{u_1, u_2, u_3\}$, $F = \{v_1, v_2\}$, 其中

$$u_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \ u_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \ u_3 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \quad \text{and} \quad v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \ v_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

定义线性变换 $T:\mathbb{R}^3\to\mathbb{R}^2$ 如下

$$T\left(\begin{bmatrix} x_1\\ x_2\\ x_3\end{bmatrix}\right)=\begin{bmatrix} 2x_2\\ -x_1\end{bmatrix}.$$

求 T 在 E 和 F 这两组有序基下的矩阵表示 A.

6. (6 points) Let A, B be $n\times n$ matrices. Suppose A and B are both symmetric. Is AB necessarily symmetric? If yes, please give a proof. Otherwise please give a counterexample.

(6 分) 设 A, B 均为 $n\times n$ 矩阵. 假设 A 和 B 都是对称矩阵. AB 是否一定是对称矩阵? 若是, 请给出证明. 否则请给出一个反例. $A=\begin{pmatrix}1&1\\1&0\end{pmatrix}$ $B=\begin{pmatrix}0&1\\1&1\end{pmatrix}$ $AB=\begin{pmatrix}1&2\\0&1\end{pmatrix}$

7. (16 points) The following two questions are independent:

(a) Let A be the 2×2 matrix such that the linear transformation $\mathbb{R}^2\to\mathbb{R}^2$, $v\mapsto Av$ rotate every vector in $\mathbb{R}^2$ through 60° counter-clockwise (about the origin). Find A and A^{2020}.

$$A=\begin{pmatrix}\frac{1}{2} & -\frac{\sqrt{3}}{2}\\ \frac{\sqrt{3}}{2} & \frac{1}{2}\end{pmatrix}=\begin{pmatrix}\cos\frac{\pi}{3} & -\sin\frac{\pi}{3}\\ \sin\frac{\pi}{3} & \cos\frac{\pi}{3}\end{pmatrix}$$

(b) Three planes Π_1, Π_2, Π_3 in the space $\mathbb{R}^3$ are given by the equations

$$\begin{aligned}\Pi_1 &: \quad x+y+z=0,\\ \Pi_2 &: \quad 2x-y+4z=0,\\ \Pi_3 &: \quad -x+2y-z=0.\end{aligned}$$

Determine a matrix representative (in the standard basis of $\mathbb{R}^3$) of a linear transformation taking the xy plane to Π_1, the yz plane to Π_2 and the zx plane to Π_3.

$$A^{2020}=\begin{pmatrix}-\frac{1}{2} & \frac{\sqrt{3}}{2}\\ -\frac{\sqrt{3}}{2} & -\frac{1}{2}\end{pmatrix}=\begin{pmatrix}\cos\frac{4\pi}{3} & -\sin\frac{4\pi}{3}\\ \sin\frac{4\pi}{3} & \cos\frac{4\pi}{3}\end{pmatrix}$$

(16 分) 以下两个小题是相互独立的:

(a) 设 A 是 2×2 矩阵使得线性变换 $\mathbb{R}^2\to\mathbb{R}^2$, $v\mapsto Av$ 把 $\mathbb{R}^2$ 中每个向量 (绕原点) 逆时针转动 60°.
求 A 和 A^{2020}.

(b) 在空间 $\mathbb{R}^3$ 中由以下方程给出三个平面 Π_1, Π_2, Π_3:

$$\begin{aligned}\Pi_1 &: \quad x+y+z=0,\\ \Pi_2 &: \quad 2x-y+4z=0,\\ \Pi_3 &: \quad -x+2y-z=0.\end{aligned}$$

求一个矩阵, 使它 (在 $\mathbb{R}^3$ 的标准基下) 表示的线性变换将 xy 平面映射成 Π_1, 将 yz 平面映射成 Π_2 并将 zx 平面映射成 Π_3.

例如. $\begin{pmatrix}-1 & -5 & -7\\ 0 & 2 & -2\\ 1 & 3 & 3\end{pmatrix}$

每一列可替换为它的任意非零常数倍

8. (12 points) Let A be a 3×3 matrix such that $\operatorname{rank}(A)=2$ and $A^3=0$.

(a) Prove that $\operatorname{rank}(A^2)=1$.

(b) Let $\alpha_1\in\mathbb{R}^3$ be a nonzero vector such that $A\alpha_1=0$. Prove that there exist vectors α_2, α_3 such that $A\alpha_2=\alpha_1$, $A^2\alpha_3=\alpha_1$.

(c) For any vectors α_2,α_3 described as above, show that $\alpha_1,\alpha_2,\alpha_3$ are linearly independent.

(*In this problem, you are allowed to assume the statements of some questions to answer subsequent questions.*)

(12 分) 设 A 是 3×3 矩阵, 它满足 $\operatorname{rank}(A)=2$ 及 $A^3=0$.

(a) 证明 $\operatorname{rank}(A^2)=1$.

(b) 设 $\alpha_1\in\mathbb{R}^3$ 是满足 $A\alpha_1=0$ 的非零向量. 证明: 存在向量 α_2, α_3 使得 $A\alpha_2=\alpha_1$, $A^2\alpha_3=\alpha_1$.

(c) 证明: 对于任意满足上述条件的向量 α_2,α_3, 向量组 $\alpha_1,\alpha_2,\alpha_3$ 线性无关.

(本题中, 允许承认前面小题的结果来用于后续问题的解答.)

(a)
$$\left.\begin{array}{l} A^3=0 \Rightarrow C(A^2)\subseteq N(A) \\ \operatorname{rank}(A)=2 \Rightarrow \dim N(A)=1 \end{array}\right\} \Rightarrow \dim C(A^2)=\operatorname{rank}(A^2)\le 1$$

If $A^2=0$ then $C(A)\subseteq N(A)$. Then $\dim C(A)=\operatorname{rank}(A)\le \dim N(A)=1$

contradiction to $\operatorname{rank}(A)=2$

So $0<\operatorname{rank}(A^2)\le 1$, $\operatorname{rank}(A^2)=1$.

(b) We have $C(A^2)\subseteq N(A)$ and $N(A)=\operatorname{Span}(\alpha_1)$.

By (a) + Dimension count $\Rightarrow$ $C(A^2)=\operatorname{span}(\alpha_1)$

$\Rightarrow \exists\ \alpha_3$ s.t. $A^2\alpha_3=\alpha_1$

Then take $\alpha_2=A\alpha_3$.

(c)
$$x_1\alpha_1+x_2\alpha_2+x_3\alpha_3=0$$

$\Rightarrow A^2(x_1\alpha_1+x_2\alpha_2+x_3\alpha_3)=0 \Rightarrow x_1\alpha_3=0$

$\Rightarrow \underline{x_1=0}$

南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

考试科目: 线 性 代 数 A　　开课单位: 数 学 系
考试时长: 120 分钟　　命题教师: 线性代数教学团队

题 号	1	2	3	4	5	6	7	8
分 值	15 分	25 分	12 分	10 分	12 分	12 分	8 分	6 分

本试卷共 (8) 大题, 满分 (100) 分. 请将所有答案写在答题本上.

This exam includes 8 questions and the score is 100 in total. Write all your answers on the examination book.

本套试卷为 B 卷 Version B

1. (15 points, 3 points each) Multiple Choice. Only one choice is correct.

(共 15 分, 每小题 3 分) 选择题, 只有一个选项是正确的.

(1) Let $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 0 & 1 & 1 \end{bmatrix}$ and let $\alpha_1, \alpha_2, \alpha_3$ be linearly independent column vectors in $\mathbb{R}^3$. Then the rank of the vector system $A\alpha_1, A\alpha_2, A\alpha_3$ ( )

(A) must be 1.

(B) must be 2.

(C) must be 3.

(D) can be 1 or 2.

设 $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 0 & 1 & 1 \end{bmatrix}$, $\alpha_1, \alpha_2, \alpha_3$ 为 $\mathbb{R}^3$ 中线性无关的向量组. 则向量组 $A\alpha_1, A\alpha_2, A\alpha_3$ 的秩 ( )

(A) 一定是 1.

(B) 一定是 2.

(C) 一定是 3.

(D) 可能是 1 也可能是 2.

(2) Let I_n be the identity matrix of order n and let α be a column vector of length 1 in $\mathbb{R}^n$. Then ( )

(A) $I_n - \alpha\alpha^T$ is not invertible.

(B) $I_n + \alpha\alpha^T$ is not invertible.

设 $\alpha_1, \alpha_2, \alpha_3$ 为 $\mathbb{R}^3$ 中的列向量, 使得矩阵 $P = (\alpha_1, \alpha_2, \alpha_3)$ 可逆. 假设 A 为 3×3 矩阵, 满足 $P^{-1}AP = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$. 则 $A(\alpha_1 + \alpha_2 + \alpha_3) = (\quad)$

(A) $\alpha_1 + \alpha_2$.

(B) $\alpha_2 + 2\alpha_3$.

(C) $\alpha_2 + \alpha_3$.

(D) $\alpha_1 + 2\alpha_3$.

(5) Which of the following statements is true? (　　)

(A) Let A be an $m\times n$ complex matrix and $B = A^H A$. (Here $A^H = \overline{A}^T$ denotes the conjugate transpose of A.) Then the matrix $B + iI_n$ is invertible.

(B) If A is a real square matrix of order n and $x^T Ax = 0$ for all $x \in \mathbb{R}^n$, then A must be the zero matrix.

(C) Suppose A and B are invertible square matrices. If A and B are similar to each other, then $A + A^T$ is similar to $B + B^T$.

(D) If A is an upper triangular (square) matrix and A is similar to a diagonal matrix, then A must be a diagonal matrix.

下列哪个论断是正确的? (　　)

(A) 设 A 为 $m\times n$ 复矩阵, $B = A^H A$. (这里 $A^H = \overline{A}^T$ 表示 A 的共轭转置.) 则矩阵 $B + iI_n$ 可逆.

(B) 如果 A 是 n 阶实方阵, 且对任意 $x \in \mathbb{R}^n$ 均有 $x^T Ax = 0$, 则 A 一定是零矩阵.

(C) 假设 A 和 B 都是可逆方阵. 如果 A 和 B 相似, 则 $A + A^T$ 和 $B + B^T$ 相似.

(D) 若 A 是上三角(方)阵并且 A 相似于某个对角阵, 则 A 一定是对角阵.

2. (25 points, 5 points each) Fill in the blanks. (共 25 分, 每小题 5 分) 填空题.

(1) Let A be a 5×4 matrix of rank 3. Then the dimension of its left null space $N(A^T)$ is __________

设 A 为秩等于 3 的 5×4 矩阵. 则它的左零空间 $N(A^T)$ 维数是 __________

(2) Let A be a square matrix of order 3 and let $\alpha_1, \alpha_2, \alpha_3$ be linearly independent vectors in $\mathbb{R}^3$. Suppose $A\alpha_1 = 2\alpha_1 + \alpha_2 + \alpha_3$, $A\alpha_2 = \alpha_2 + 2\alpha_3$ and $A\alpha_3 = -\alpha_2 + \alpha_3$. Then the real eigenvalues of A are __________

设 A 为 3 阶方阵, $\alpha_1, \alpha_2, \alpha_3$ 为 $\mathbb{R}^3$ 中的线性无关向量组. 假设 $A\alpha_1 = 2\alpha_1 + \alpha_2 + \alpha_3$, $A\alpha_2 = \alpha_2 + 2\alpha_3$ 且 $A\alpha_3 = -\alpha_2 + \alpha_3$. 则 A 的实特征值为 __________

(3) Let A, B be square matrices of order n. Suppose $|A| = 3$, $|B| = 2$ and $|A^{-1} + B| = 2$. Then $|A + B^{-1}| =$__________

设 A, B 为 n 阶方阵. 假设 $|A| = 3$, $|B| = 2$ 而 $|A^{-1}+B| = 2$. 则 $|A+B^{-1}| =$__________

(a) Find a recurrence relation relating $D_n(x, y)$ to $D_{n-1}(x, y)$ and $D_{n-2}(x, y)$ for $n \geq 4$.

(b) Compute the determinant $D_n(x, y)$ for all $n \geq 2$.

(10 分) 设 $x, y \in \mathbb{R}$, $n \geq 2$. 考虑以下 n 阶行列式:

$$D_n(x, y) = \begin{vmatrix} x+y & xy & 0 & \dots & 0 & 0 \\ 1 & x+y & xy & \dots & 0 & 0 \\ 0 & 1 & x+y & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 1 & x+y \end{vmatrix}$$

当 $n=2$ 时, 我们认为 $D_2(x, y)$ 是 $\begin{vmatrix} x+y & xy \\ 1 & x+y \end{vmatrix}$.

(a) 在 $n \geq 4$ 时找出能将 $D_n(x, y)$ 和 $D_{n-1}(x, y), D_{n-2}(x, y)$ 建立联系的一个递推关系式.

(b) 对任意 $n \geq 2$ 计算行列式 $D_n(x, y)$.

5. (12 points) Let $a, b \in \mathbb{R}$ and put

$$A = \begin{bmatrix} 2 & -1 & 2 \\ 5 & a & 3 \\ -1 & b & -2 \end{bmatrix}.$$

Suppose that $p = (1, 1, -1)^T$ is an eigenvector of A.

(a) Find the values of a, b and find the eigenvalue λ corresponding to the eigenvector p.

(b) Is the matrix A diagonalizable? Please explain your answer.

(12 分) 设 $a, b \in \mathbb{R}$, 令

$$A = \begin{bmatrix} 2 & -1 & 2 \\ 5 & a & 3 \\ -1 & b & -2 \end{bmatrix}.$$

假设 $p = (1, 1, -1)^T$ 是 A 的一个特征向量.

(a) 求 a, b 的值并找出特征向量 p 对应的特征值 λ.

(b) 矩阵 A 是否可对角化? 请解释你的答案.

6. (12 points) Consider the ternary quadratic form

$$f(x_1, x_2, x_3) = (x_1 - x_2 + x_3)^2 + (x_2 + x_3)^2 + (x_1 + ax_3)^2$$

where $a \in \mathbb{R}$ is a parameter.

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考试科目: 线 性 代 数 A　　开课单位: 数 学 系
考试时长: 120 分钟　　命题教师: 线性代数教学团队

题 号	1	2	3	4	5	6	7	8
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本试卷共 (8) 大题, 满分 (100) 分. 请将所有答案写在答题本上.

This exam includes 8 questions and the score is 100 in total. Write all your answers on the examination book.

本套试卷为 B 卷 Version B

1. (15 points, 3 points each) Multiple Choice. Only one choice is correct.

(共 15 分, 每小题 3 分) 选择题, 只有一个选项是正确的.

(1) Let $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 0 & 1 & 1 \end{bmatrix}$ and let $\alpha_1, \alpha_2, \alpha_3$ be linearly independent column vectors in $\mathbb{R}^3$. Then the rank of the vector system $A\alpha_1, A\alpha_2, A\alpha_3$ ( )

(A) must be 1.

(B) must be 2.

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(D) can be 1 or 2.

设 $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 0 & 1 & 1 \end{bmatrix}$, $\alpha_1, \alpha_2, \alpha_3$ 为 $\mathbb{R}^3$ 中线性无关的向量组. 则向量组 $A\alpha_1, A\alpha_2, A\alpha_3$ 的秩 ( )

(A) 一定是 1.

(B) 一定是 2.

(C) 一定是 3.

(D) 可能是 1 也可能是 2.

(2) Let I_n be the identity matrix of order n and let α be a column vector of length 1 in $\mathbb{R}^n$. Then ( )

(A) $I_n - \alpha\alpha^T$ is not invertible.

(B) $I_n + \alpha\alpha^T$ is not invertible.

设 $\alpha_1, \alpha_2, \alpha_3$ 为 $\mathbb{R}^3$ 中的列向量, 使得矩阵 $P = (\alpha_1, \alpha_2, \alpha_3)$ 可逆. 假设 A 为 3×3 矩阵, 满足 $P^{-1}AP = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$. 则 $A(\alpha_1+\alpha_2+\alpha_3) = (\quad)$

(A) $\alpha_1+\alpha_2$.

(B) $\alpha_2+2\alpha_3$.

(C) $\alpha_2+\alpha_3$.

(D) $\alpha_1+2\alpha_3$.

(5) Which of the following statements is true? ( )

(A) Let A be an $m\times n$ complex matrix and $B = A^HA$. (Here $A^H = \overline{A}^T$ denotes the conjugate transpose of A.) Then the matrix $B + iI_n$ is invertible.

(B) If A is a real square matrix of order n and $x^TAx = 0$ for all $x \in \mathbb{R}^n$, then A must be the zero matrix.

(C) Suppose A and B are invertible square matrices. If A and B are similar to each other, then $A + A^T$ is similar to $B + B^T$.

(D) If A is an upper triangular (square) matrix and A is similar to a diagonal matrix, then A must be a diagonal matrix.

下列哪个论断是正确的? ( )

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(B) 如果 A 是 n 阶实方阵, 且对任意 $x \in \mathbb{R}^n$ 均有 $x^TAx = 0$, 则 A 一定是零矩阵.

(C) 假设 A 和 B 都是可逆方阵. 如果 A 和 B 相似, 则 $A + A^T$ 和 $B + B^T$ 相似.

(D) 若 A 是上三角(方)阵并且 A 相似于某个对角阵, 则 A 一定是对角阵.

Solution. (1) B (2) A (3) C (4) B (5) D.

2. (25 points, 5 points each) Fill in the blanks. (共 25 分, 每小题 5 分) 填空题.

(1) Let A be a 5×4 matrix of rank 3. Then the dimension of its left null space $N(A^T)$ is ________

设 A 为秩等于 3 的 5×4 矩阵. 则它的左零空间 $N(A^T)$ 维数是 ________

(2) Let A be a square matrix of order 3 and let $\alpha_1, \alpha_2, \alpha_3$ be linearly independent vectors in $\mathbb{R}^3$. Suppose $A\alpha_1 = 2\alpha_1+\alpha_2+\alpha_3$, $A\alpha_2 = \alpha_2+2\alpha_3$ and $A\alpha_3 = -\alpha_2+\alpha_3$. Then the real eigenvalues of A are ________

设 A 为 3 阶方阵, $\alpha_1, \alpha_2, \alpha_3$ 为 $\mathbb{R}^3$ 中的线性无关向量组. 假设 $A\alpha_1 = 2\alpha_1+\alpha_2+\alpha_3$, $A\alpha_2 = \alpha_2+2\alpha_3$ 且 $A\alpha_3 = -\alpha_2+\alpha_3$. 则 A 的实特征值为 ________

(3) Let A, B be square matrices of order n. Suppose $|A| = 3$, $|B| = 2$ and $|A^{-1}+B| = 2$. Then $|A+B^{-1}| =$________

设 A, B 为 n 阶方阵. 假设 $|A| = 3$, $|B| = 2$ 而 $|A^{-1}+B| = 2$. 则 $|A+B^{-1}| =$________

(b)

$$Q = \begin{bmatrix} \frac{1}{5} & -\frac{2}{5} & -\frac{4}{5} \\ \frac{2}{5} & \frac{1}{5} & \frac{2}{5} \\ \frac{2}{5} & -\frac{4}{5} & \frac{2}{5} \\ \frac{4}{5} & \frac{2}{5} & -\frac{1}{5} \end{bmatrix}, \ R = \begin{bmatrix} 5 & -2 & 1 \\ 2 & 0 & 1 \\ 0 & 0 & 2 \end{bmatrix}.$$

4. (10 points) Let $x, y \in \mathbb{R}$, $n \geq 2$ and consider the following determinant of order n:

$$D_n(x, y) = \begin{vmatrix} x+y & xy & 0 & \dots & 0 & 0 \\ 1 & x+y & xy & \dots & 0 & 0 \\ 0 & 1 & x+y & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 1 & x+y \end{vmatrix}$$

For $n = 2$, we consider $D_2(x, y)$ as $\begin{vmatrix} x+y & xy \\ 1 & x+y \end{vmatrix}$.

(a) Find a recurrence relation relating $D_n(x, y)$ to $D_{n-1}(x, y)$ and $D_{n-2}(x, y)$ for $n \geq 4$.

(b) Compute the determinant $D_n(x, y)$ for all $n \geq 2$.

(10 分) 设 $x, y \in \mathbb{R}$, $n \geq 2$. 考虑以下 n 阶行列式:

$$D_n(x, y) = \begin{vmatrix} x+y & xy & 0 & \dots & 0 & 0 \\ 1 & x+y & xy & \dots & 0 & 0 \\ 0 & 1 & x+y & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 1 & x+y \end{vmatrix}$$

当 $n = 2$ 时, 我们认为 $D_2(x, y)$ 是 $\begin{vmatrix} x+y & xy \\ 1 & x+y \end{vmatrix}$.

(a) 在 $n \geq 4$ 时找出能将 $D_n(x, y)$ 和 $D_{n-1}(x, y), D_{n-2}(x, y)$ 建立联系的一个递推关系式.

(b) 对任意 $n \geq 2$ 计算行列式 $D_n(x, y)$.

Solution.

(a)

$$D_n = (x+y)D_{n-1} - xyD_{n-2}.$$

(b) $D_n(x, y) = x^n + x^{n-1}y + x^{n-2}y^2 + \cdots + x^2y^{n-2} + xy^{n-1} + y^n$.

5. (12 points) Let $a, b \in \mathbb{R}$ and put

$$A = \begin{bmatrix} 2 & -1 & 2 \\ 5 & a & 3 \\ -1 & b & -2 \end{bmatrix}.$$

Suppose that $p = (1, 1, -1)^T$ is an eigenvector of A.

Solution.

(a) $a \neq -2$.

(b) $a = 2$.

(c)

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & 2 \\ 1 & 0 & 2 \end{bmatrix}$$

7. (8 points) Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & -1 \end{bmatrix}$.

   (a) Find all the singular values of A.

   (b) Find the singular value decomposition of A, in other words, find two orthogonal matrices U and V (of suitable size) such that $A = U\Sigma V^T$.

   (8 分) 令 $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & -1 \end{bmatrix}$.

   (a) 求 A 的所有奇异值.

   (b) 求 A 的奇异值分解. 即, 找出两个 (适当大小的) 正交矩阵 U 和 V 使得 $A = U\Sigma V^T$.

   Solution.

   (a) The singular values of A are 2, $\sqrt{2}$.

   (b)

$$U = \begin{bmatrix} -\frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \end{bmatrix}, V = \begin{bmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix}, \Sigma = \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix}.$$

8. (6 points) Find a real 4×4 orthogonal matrix A such that A has no real eigenvalues but both A^2 and A^3 have real eigenvalues. Please explain why the matrix you give has the required properties.

   (6 分) 给出一个 4×4 实正交矩阵 A 使得 A 没有实特征值, 而 A^2 和 A^3 都有实特征值. 请解释为什么你给出的矩阵满足要求.

   Solution.

$$A = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Solutions

南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

考试科目: 线性代数 A　　开课单位: 数学系
考试时长: 120 分钟　　命题教师: 线性代数教学团队

题 号	1	2	3	4	5	6	7	8
分 值	15 分	25 分	10 分	16 分	10 分	6 分	16 分	12 分

本试卷共 (8) 大题, 满分 (110) 分. 请将所有答案写在答题本上.

This exam includes 8 questions and the score is 110 in total. Write all your answers on the examination book.

1. (15 points, 3 points each) Multiple Choice. Only one choice is correct.

(共 15 分, 每小题 3 分) 选择题, 只有一个选项是正确的.

(1) Let A be an $m \times n$ real matrix and b be a column vector in $\mathbb{R}^m$. Which of the following statements is correct?

A

(A) If $Ax = b$ has infinite many solutions, then $Ax = 0$ has a nonzero solution.

(B) If the system $Ax = 0$ has only zero solution, then $Ax = b$ has one and only one solution.

(C) If the rank of A is n, then the system $Ax = b$ must have a solution.

(D) If A is a square matrix (i.e., $m = n$), then the system $Ax = b$ is consistent if and only if A is invertible.

设 A 为 $m \times n$ 实矩阵, b 是 $\mathbb{R}^m$ 中的列向量. 下列陈述中哪个是正确的？

(A) 如果 $Ax = b$ 有无穷多个解, 则 $Ax = 0$ 有非零解.

(B) 如果方程组 $Ax = 0$ 只有零解, 则 $Ax = b$ 有且仅有一个解.

(C) 如果 A 的秩为 n, 则方程组 $Ax = b$ 必有解.

(D) 如果 A 是方阵 (即 $m = n$), 则方程组 $Ax = b$ 是相容的当且仅当 A 可逆.

C

(2) Suppose A is an $m \times n$ matrix, B is an $n \times m$ matrix, and I is the $m \times m$ identity matrix. If $AB = I$, then (　　)

(A) the column vectors of A are linearly independent, and the row vectors of B are linearly independent.

(B) the column vectors of A are linearly independent, and the column vectors of B are linearly independent.

(C) the row vectors of A are linearly independent, and the column vectors of B are linearly independent.

(C) $t \neq 0$.

(D) t 可以是任意实数.

C

(5) Which of the following statements is incorrect? ( )

(A) For any matrix A, $\operatorname{rank}(A) = \dim C(A)$.

(B) If $v_1, \cdots, v_m$ are pairwise orthogonal nonzero vectors, then the vectors $v_1, \cdots, v_m$ are linear independent.

(C) If A is an upper triangular $n \times n$ matrix such that $A^2 = 0$, then $A = 0$.

(D) Let A, B be $n \times n$ matrices such that AB is invertible. Then both A and B are invertible.

下列哪个论断是错误的? ( )

(A) 对于任意矩阵 A, $\operatorname{rank}(A) = \dim C(A)$.

(B) 如果 $v_1, \cdots, v_m$ 是一组两两正交的非零向量, 则向量组 $v_1, \cdots, v_m$ 线性无关.

(C) 如果 A 是 $n \times n$ 上三角矩阵且 $A^2 = 0$, 则 $A = 0$.

(D) 设 A, B 为 $n \times n$ 矩阵且 AB 可逆. 则 A 和 B 都可逆.

2. (25 points, 5 points each) Fill in the blanks. (共 25 分, 每小题 5 分) 填空题.

(1) Let A, B be invertible $n \times n$ matrices. Then the inverse of the block matrix $\begin{bmatrix} 0 & A \\ B & 0 \end{bmatrix}$ is ________

设 A, B 均为 $n \times n$ 可逆矩阵. 则分块矩阵 $\begin{bmatrix} 0 & A \\ B & 0 \end{bmatrix}$ 的逆为 $\begin{bmatrix} 0 & B^{-1} \\ A^{-1} & 0 \end{bmatrix}$

(2) Suppose A is a 3×4 matrix and $\dim N(A) = 2$. Then $\dim N(A^T) =$ ________

设 A 为 3×4 矩阵且 $\dim N(A) = 2$. 则 $\dim N(A^T) =$ 1

(3) Let $A = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix}$. Then $A^{-1} =$ $\begin{bmatrix} 1 & 0 & 0 \\ -a & 1 & 0 \\ ac-b & -c & 1 \end{bmatrix}$

设 $A = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix}$. 则 $A^{-1} =$ ________

(4) Let u, v be vectors in $\mathbb{R}^n$ such that $\|u\| = 3$, $\|v\| = 4$ and $u^T v = -3$.

Then $\|2u + 3v\| =$ ________

设 u, v 为 $\mathbb{R}^n$ 中的向量, 满足 $\|u\| = 3$, $\|v\| = 4$ 以及 $u^T v = -3$. 则 $\|2u+3v\| =$ 12

(5) Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & -1 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}$.

Then the least squares solution to $Ax = b$ is $\hat{x} =$ $\begin{bmatrix} 4 \\ -5 \end{bmatrix}$

定义线性变换 $T:\mathbb{R}^3\to\mathbb{R}^2$ 如下

$$T\left(\begin{bmatrix} x_1\\ x_2\\ x_3\end{bmatrix}\right)=\begin{bmatrix} 2x_2\\ -x_1\end{bmatrix}.$$

求 T 在 E 和 F 这两组有序基下的矩阵表示 A.

6. (6 points) Let A, B be $n\times n$ matrices. Suppose A and B are both symmetric. Is AB necessarily symmetric? If yes, please give a proof. Otherwise please give a counterexample.

(6 分) 设 A, B 均为 $n\times n$ 矩阵. 假设 A 和 B 都是对称矩阵. AB 是否一定是对称矩阵? 若是, 请给出证明. 否则请给出一个反例.

$A=\begin{pmatrix} 1 & 1\\ 1 & 0\end{pmatrix}$ $B=\begin{pmatrix} 0 & 1\\ 1 & 1\end{pmatrix}$ $AB=\begin{pmatrix} 1 & 2\\ 0 & 1\end{pmatrix}$

7. (16 points) The following two questions are independent:

(a) Let A be the 2×2 matrix such that the linear transformation $\mathbb{R}^2\to\mathbb{R}^2$, $v\mapsto Av$ rotate every vector in $\mathbb{R}^2$ through $60°$ counter-clockwise (about the origin). Find A and A^{2020}.

$$A=\begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2}\\ \frac{\sqrt{3}}{2} & \frac{1}{2}\end{pmatrix}=\begin{pmatrix} \cos\frac{\pi}{3} & -\sin\frac{\pi}{3}\\ \sin\frac{\pi}{3} & \cos\frac{\pi}{3}\end{pmatrix}$$

(b) Three planes Π_1, Π_2, Π_3 in the space $\mathbb{R}^3$ are given by the equations

$$\begin{aligned}\Pi_1 &: \ x+y+z=0,\\ \Pi_2 &: \ 2x-y+4z=0,\\ \Pi_3 &: \ -x+2y-z=0.\end{aligned}$$

Determine a matrix representative (in the standard basis of $\mathbb{R}^3$) of a linear transformation taking the xy plane to Π_1, the yz plane to Π_2 and the zx plane to Π_3.

$$A^{2020}=\begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2}\\ -\frac{\sqrt{3}}{2} & -\frac{1}{2}\end{pmatrix}=\begin{pmatrix} \cos\frac{4\pi}{3} & -\sin\frac{4\pi}{3}\\ \sin\frac{4\pi}{3} & \cos\frac{4\pi}{3}\end{pmatrix}$$

(16 分) 以下两个小题是相互独立的:

(a) 设 A 是 2×2 矩阵使得线性变换 $\mathbb{R}^2\to\mathbb{R}^2$, $v\mapsto Av$ 把 $\mathbb{R}^2$ 中每个向量 (绕原点) 逆时针转动 $60°$.
求 A 和 A^{2020}.

(b) 在空间 $\mathbb{R}^3$ 中由以下方程给出三个平面 Π_1, Π_2, Π_3:

$$\begin{aligned}\Pi_1 &: \ x+y+z=0,\\ \Pi_2 &: \ 2x-y+4z=0,\\ \Pi_3 &: \ -x+2y-z=0.\end{aligned}$$

求一个矩阵, 使它 (在 $\mathbb{R}^3$ 的标准基下) 表示的线性变换将 xy 平面映射成 Π_1, 将 yz 平面映射成 Π_2 并将 zx 平面映射成 Π_3.

例如 $\begin{pmatrix} -1 & -5 & -7\\ 0 & 2 & -2\\ 1 & 3 & 3\end{pmatrix}$

每一列可替换为它的任意非零常数倍